EE361 Electromechanical Energy Conversion I — METU EEE — full-semester study notes

Chapter III — Transformers

Transformatörler · Weeks 7–9 of the 2026 syllabus

- Lesson 17 — The ideal transformer
- Lesson 18 — The practical transformer and its equivalent circuits
- Lesson 19 — Transformer tests: open-circuit and short-circuit
- Lesson 20 — Transformer losses, efficiency and voltage regulation
- Lesson 21 — Three-phase transformers
- Chapters II–III — remaining archive problems with full solutions

> Assistant-written teaching notes (English with Turkish terminology). Teacher sources LN7–LN11; textbook Sen Ch. 2. Appendix: remaining Chapter II–III archive problems with full solutions. Every numerical result is checked by `assistant/calculations/full-semester/run_all.py`; source pages are cited in each lesson. Printed 2026-11-02 from the Markdown sources in `assistant/lessons/`.

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Chapters II–III — remaining archive problems with full solutions

1. MT I Fall 2015 Q2 — trapezoidal flux on a hysteretic core
2. MT I Fall 2015 Q3(a)–(c) — voltage, frequency and saturation
3. MT I Fall 2019 Q2(e) — design an ungapped inductor with the same L and saturation current
4. MT I Fall 2019 Q3 — 1.25 MVA, 34.5 kV / 230 V distribution transformer
5. Final Fall 2016 Q1 — 1 MVA, 33 kV / 400 V transformer in per unit
6. Final Fall 2017 Q2 Part I — EMF equation and power rating
7. MT1 recitation (Dr. Ermiş) and Homework 1

Lesson 17 — The ideal transformer

Chapter III: Transformers (Transformatörler) · Week 7 · Lesson 17 of 31

Assistant-written teaching note. Teacher source: LN7 (all 22 slides). Book: Sen 3rd ed., chapter opening and §2.1 (printed pp. 40–48, viewer 64–72), Example 2.1, ratings (p. 51). Links: Lessons 14–15 (mutual inductance, Faraday's law); 2025 MT1 Q4(d). Numbers: `lesson17_ideal_transformer.py`.

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1. Place and goals

A transformer is "a magnetically coupled circuit, usually with a highly permeable core" (LN7 slide 20). It is not an electromechanical converter — nothing moves — but it is indispensable: generators produce power at about 10–30 kV, it is transmitted at hundreds of kV to keep currents and losses low (Chapter I, Lesson 2), and it is used at 230/400 V. Every change of voltage level is a transformer. Chapter III develops the transformer model step by step: the **ideal transformer** here, its non-idealities and equivalent circuit (Lesson 18), how to measure the parameters (Lesson 19), its performance — efficiency and voltage regulation (Lesson 20) — and three-phase connections (Lesson 21).

After this lesson you should be able to:

1. State the ideal-transformer assumptions and derive the voltage, current and power relations from Faraday's and Ampère's laws.
2. Use dot (polarity) marks correctly in circuits and phasor diagrams, and describe the polarity test.
3. Refer voltages, currents and impedances from one side to the other, and use impedance transformation for matching.
4. Interpret nameplate ratings and compute rated currents.
5. Connect the ideal transformer to the coupled-coil model of Lesson 14.

Prerequisites: Lesson 14 (mutual inductance, dots), Lesson 15 (Faraday, $E = 4.44 f N \Phi_{max}$), Chapter I phasors and complex power.

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2. Construction in one paragraph

Two (or more) windings are placed on a common laminated steel core (Sen Fig. 2.1). In the **core type** the windings sit on the two legs of a rectangular core; in the **shell type** both windings sit on the centre leg of a three-legged core. Laminations (0.3–0.5 mm, Lesson 16) are stacked with alternating joints so that no continuous air gap exists, because a gap would demand a large magnetizing current (Lesson 13). To keep leakage small the windings are wound one over the other (LV winding nearest the core, HV outside) or interleaved (LN8 slides 10–11). The winding connected to the source is the **primary** (birincil sargı); the one connected to the load is the **secondary** (ikincil sargı). Independently of that, the one with more turns is the **high-voltage (HV)** winding and the other the **low-voltage (LV)** winding.

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3. The ideal transformer

3.1 Assumptions (LN7 slide 21; Sen §2.1)

1. Winding resistances are zero.
2. The core is infinitely permeable ($\mu \to \infty$), so zero net MMF is needed to set up any flux.
3. No leakage flux: all flux is confined to the core and links both windings.
4. No core losses (no hysteresis, no eddy currents).
5. Inter-winding and stray capacitances are neglected.

3.2 Voltage ratio

Apply v_1 to winding 1 (N_1 turns). By assumption 3 the same core flux Φ links both windings (Figure 17.1). With assumption 1 and the passive sign convention on winding 1 (Lesson 15, Eq. 15.2):

$$v_1 = e_1 = N_1 \frac{d\Phi}{dt}, \qquad v_2 = e_2 = N_2 \frac{d\Phi}{dt}$$

(LN7 slide 3; Sen Eqs. 2.1–2.2), where v_2 is measured positive at the terminal that is dotted like winding 1's positive terminal. Dividing:

$$\boxed{\frac{v_1}{v_2} = \frac{N_1}{N_2} = a} \qquad \text{(turns ratio, LN7 slide 7: } a = N_1/N_2\text{).} \tag{17.1}$$

The ratio holds instant by instant, for any waveform for which the flux can follow — but not for DC: a constant v_1 would require an ever-growing flux (Lesson 15, Eq. 15.3), which no real core can provide. "Higher the number of turns, higher the induced voltage" (LN7 slide 4).

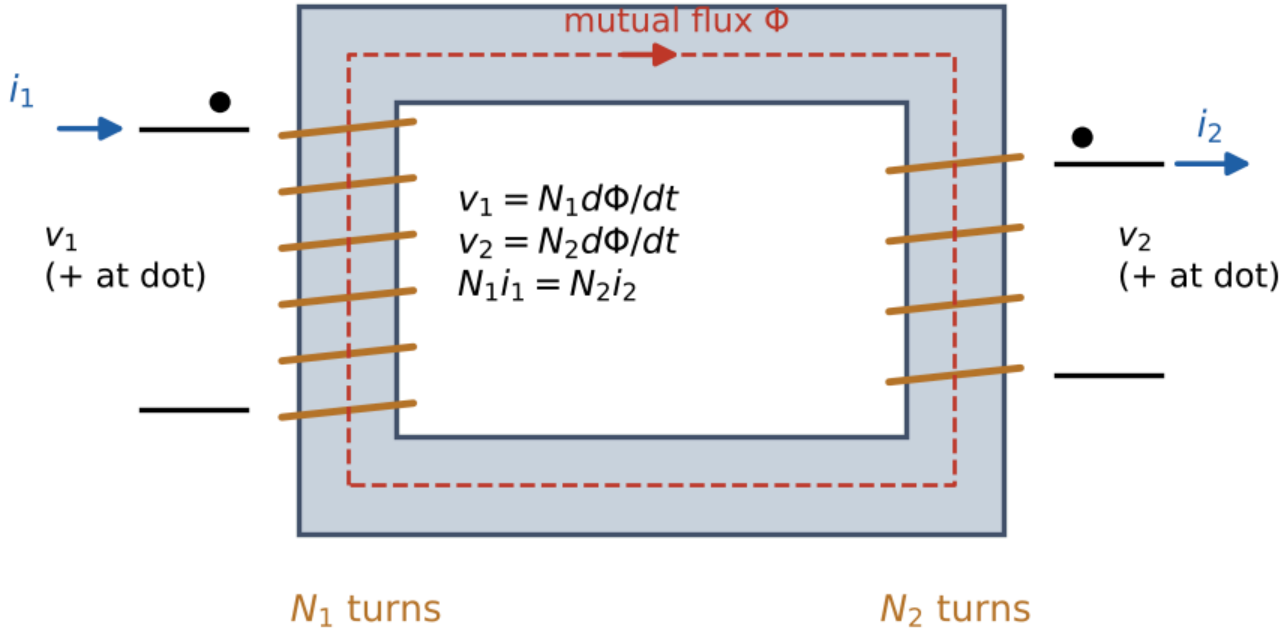


Figure 17.1. Ideal transformer: the mutual flux Φ links both windings. Dots mark like-polarity terminals; i_1 enters the dotted primary terminal and i_2 leaves the dotted secondary terminal (the convention of LN7 slides 7–9).

3.3 Current ratio from MMF balance

Connect a load: current i_2 flows out of the dotted secondary terminal. Its MMF $N_2 i_2$ tends to *reduce* the core flux (Lenz). But the flux is fixed by v_1 through (17.1); if it changed, e_1 would no longer balance v_1. The primary therefore immediately draws a current i_1 whose MMF cancels $N_2 i_2$. With $\mu \to \infty$ the net MMF around the core must be zero (Ampère's law, $\oint H\,dl = 0$ because $H = B/\mu \to 0$):

$$N_1 i_1 - N_2 i_2 = 0 \quad \Longrightarrow \quad \boxed{\frac{i_1}{i_2} = \frac{N_2}{N_1} = \frac{1}{a}} \tag{17.2}$$

(LN7 slide 6: "also valid from the magnetic circuit: MMF on both sides are equal"; Sen Eqs. 2.4–2.6). Sen's remark is worth memorizing: *it is this MMF-balancing requirement that makes the primary "know" about the current in the secondary.* There is no electrical connection, yet load current is drawn from the supply.

3.4 Power balance

Multiplying (17.1) and (17.2):

$$v_1 i_1 = v_2 i_2 \qquad \text{and, for sinusoids,} \quad \mathbf{V}_1 \mathbf{I}_1^{*} = \mathbf{V}_2 \mathbf{I}_2^{*} \;\; (S_1 = S_2). \tag{17.3}$$

Instantaneous input power equals output power (LN7 slide 5: $P_1 = P_2$, $V_1 I_1 = V_2 I_2$). Real *and* reactive power pass through unchanged; an ideal transformer neither consumes nor generates vars. It changes the *level* of voltage and current, keeping their product.

3.5 Phasors

With sinusoidal excitation, RMS phasors obey $\mathbf{V}_1 = a\mathbf{V}_2$ and $\mathbf{I}_1 = \mathbf{I}_2/a$ (Sen Eqs. 2.8–2.9): primary and secondary voltages are **in phase** when measured between like-polarity terminals, and so are the currents (with the convention i_1 into a dot, i_2 out of a dot).

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4. Polarity (dot) marks

Terminals 1 (primary) and 3 (secondary) are **like-polarity** (aynı polariteli) terminals if currents entering them produce flux in the same direction in the core; then e_{12} and e_{34} are in phase (Sen §2.1.2). They are marked with dots or ±. LN7 slide 8 warns: "Beware of dots" — reversing a winding's connection reverses the sign of its voltage relative to the other.

Polarity test (Sen Fig. 2.8c). Connect one terminal of each winding together (2 to 4), energize winding 1–2 at reduced voltage and measure V_{12}, V_{34} and V_{13}:

- $V_{13} \approx V_{12} + V_{34}$ ("additive"): terminals 1 and **4** are like-polarity;
- $V_{13} \approx V_{12} - V_{34}$ ("subtractive"): terminals 1 and **3** are like-polarity.

Why it matters: transformers in parallel must be connected with like terminals together, otherwise their EMFs add around the loop and drive a huge circulating current (Sen Fig. 2.9; Example 17.6). Three-phase connections (Lesson 21) are impossible to build correctly without polarity marks.

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5. Referring quantities and impedance transformation

5.1 Impedance seen through an ideal transformer

Put an impedance Z_2 on the secondary: $\mathbf{V}_2 = Z_2\mathbf{I}_2$. Seen from the primary,

$$Z_{in} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{a\mathbf{V}_2}{\mathbf{I}_2/a} = a^2 Z_2 \qquad \text{(Sen Eqs. 2.11–2.12).} \tag{17.4}$$

5.2 Referring rules (LN7 slides 10–11)

A whole secondary circuit can be **referred** (yansıtılmış) to the primary — replaced by an equivalent circuit on the primary side that draws the same primary current and power — with

$$V_2' = aV_2, \qquad I_2' = I_2/a, \qquad Z_2' = a^2 Z_2.$$

Conversely, primary quantities referred to the secondary are $V_1' = V_1/a$, $I_1' = aI_1$, $Z_1' = Z_1/a^2$. A prime always means “referred to the other side”. Power, energy, and per-unit values (Lesson 21) are unchanged by referring. After referring, the ideal transformer can be removed from the drawing (Figure 17.2); at the end of the calculation, convert the referred results back to actual values.

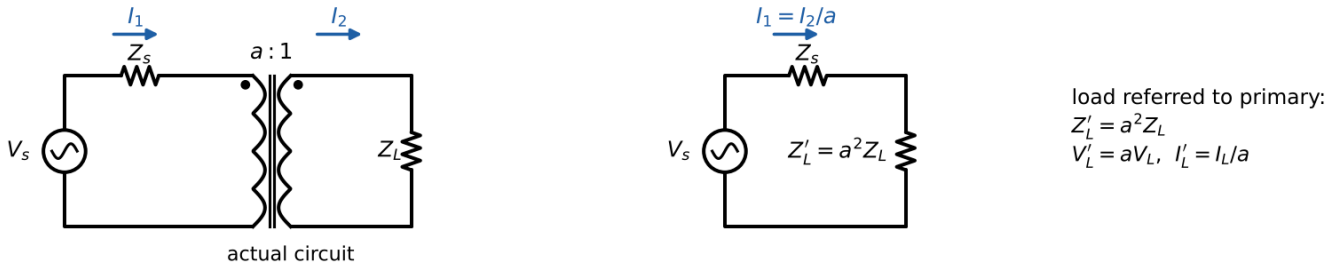


Figure 17.2. Left: actual circuit. Right: the same circuit with the load referred to the primary; the primary current and source power are identical.

Rule of thumb for which way to multiply: going to the higher-voltage side, voltages go up (×a), currents go down (÷a), impedances go up (×a²). A useful check: the impedance’s own power I^2R must come out the same on both sides.

5.3 Impedance matching

Maximum power transfer from a source with internal resistance R_s to a resistive load R_L occurs when the load “seen” by the source equals R_s. A transformer can make it so: choose a with $a^2R_L = R_s$ (Sen Example 2.1, Example 17.2). This is how audio amplifiers drove loudspeakers before modern amplifiers, and how RF and pulse transformers match impedances today.

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6. Ratings

A nameplate such as “10 kVA, 1100/110 V” means: each winding is designed for 10 kVA; the rated voltages are proportional to the turns ($a = 10$); rated currents are $10\,000/1100 = 9.09$ A (HV) and $10\,000/110 = 90.9$ A (LV) (Sen p. 51). The kVA (not kW) rating reflects that heating depends on current (copper loss) and voltage (core loss), not on power factor.

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7. From coupled coils to the ideal transformer

Lesson 14 described two coupled coils by L_1, L_2, M. For a core with no leakage ($k = 1$) and finite reluctance $\mathcal{R}$: $L_1 = N_1^2/\mathcal{R}$, $L_2 = N_2^2/\mathcal{R}$, $M = N_1 N_2/\mathcal{R}$. Substituting into $v_1 = L_1 \dot{i}_1 - M \dot{i}_2$, $v_2 = M \dot{i}_1 - L_2 \dot{i}_2$ (current i_2 *leaving* the dot) gives

$$\frac{v_1}{v_2} = \frac{N_1}{N_2} \quad \text{exactly, and} \quad i_1 = \frac{N_2}{N_1} i_2 + \underbrace{\frac{1}{L_1}\int v_1 \, dt}_{\text{magnetizing current}} \; .$$

So **perfectly coupled coils = ideal transformer + a magnetizing inductance L_1 across the primary.** Letting $\mathcal{R} \to 0$ makes $L_1 \to \infty$, the magnetizing current vanishes, and the ideal transformer remains. Example 17.5 checks this numerically. This is precisely the first step from the ideal to the practical transformer model (Lesson 18, LN8 slide 12: magnetizing reactance X_m). The 2025 MT1 Q4(d) circuit (Lesson 15, Example 15.4) is this model with $\mathcal{R} \neq 0$; its ideal-transformer limit is 0.6 A and 12 V peak.

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8. Methods and common mistakes

1. Identify which winding is primary/secondary and HV/LV; write $a = N_1/N_2$ (or the voltage ratio) explicitly.
2. Choose one side; refer everything else to it with $\times a^{\pm 1}$ for V, I and $\times a^{\pm 2}$ for Z.
3. Solve the circuit (Chapter I methods); convert back to actual quantities.
4. Check: $S_1 = S_2$ for an ideal transformer; currents inversely, voltages directly proportional to turns.

Mistakes: referring impedances with a instead of a^2; mixing referred and actual quantities in one equation; forgetting the dot convention when writing phasor equations; claiming a transformer works on DC; confusing “primary” with “HV”.

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9. Worked examples

Example 17.1 — Basic ratios

Given. An ideal 230/115 V transformer supplies a load $Z_2 = 10\angle 30^\circ\ \Omega$ from the 230 V side. **Find** V_2, I_2, I_1, the input impedance and check power balance.

Solution. $a = 2$. $V_2 = 230/2 = 115$ V. $\mathbf{I}_2 = 115/10\angle 30^\circ = 11.5\angle -30^\circ$ A. $\mathbf{I}_1 = \mathbf{I}_2/a = 5.75\angle -30^\circ$ A. $Z_{in} = 230/5.75\angle -30^\circ = 40\angle 30^\circ\ \Omega = a^2 Z_2$ ✓. $S_1 = 230 \times 5.75\angle 30^\circ = 1322.5\angle 30^\circ$ VA $= S_2 = 115 \times 11.5\angle 30^\circ$ ✓ (P = 1145 W, Q = 661 var on both sides). **Interpretation.** The transformer passes the load’s power factor unchanged to the source.

Example 17.2 — Impedance matching a loudspeaker (Sen Example 2.1)

Given. A 10 V source with 1 Ω internal resistance drives a 9 Ω speaker. **Find** the speaker power (a) directly; (b) through an ideal 1:3 transformer (source side 1).

Solution. (a) $I = 10/(1+9) = 1$ A, $P = 9$ W. (b) $a = 1/3$: the speaker referred to the source side is $a^2(9) = 1\ \Omega$. $I_1 = 10/2 = 5$ A, $P = 5^2 \times 1 = 25$ W. On the speaker side: $I_2 = aI_1 = 5/3$ A and $I_2^2 \times 9 = 25$ W ✓. **Check.** A sweep of a confirms the maximum at $a = 1/3$, where $a^2 R_L = R_s$.

Interpretation. Matching raises the delivered power from 9 W to the maximum available 25 W ($V_s^2/4R_s$), at 50 % transfer efficiency.

Example 17.3 — Solving a circuit by referring, both ways (original)

Given. A 2400 V source with series impedance $Z_s = 2 + j6\ \Omega$ feeds an ideal 10:1 transformer whose secondary supplies $Z_L = 2 + j1.5\ \Omega$. **Find** I_1, I_2, V_L and the powers.

Solution (referred to primary). $Z_L' = 100 Z_L = 200 + j150\ \Omega$. $\mathbf{I}_1 = 2400/(202 + j156) = 9.40\angle -37.7^\circ$ A. $\mathbf{I}_2 = a\mathbf{I}_1 = 94.0\angle -37.7^\circ$ A; $\mathbf{V}_L = \mathbf{I}_2 Z_L = 235.1\angle -0.81^\circ$ V. **Check (referred to secondary).** $V_s' = 240$ V, $Z_s' = 0.02 + j0.06\ \Omega$: $\mathbf{I}_2 = 240/(2.02 + j1.56)$ — the same 94.0 A. Load power $|I_2|^2(2) = 17.68$ kW; source-impedance loss $|I_1|^2(2) = 177$ W (the same whether computed as $|I_1|^2 Z_s$ or $|I_2|^2 Z_s'$). **Interpretation.** Referring is a bookkeeping choice; both sides give identical physical answers.

Example 17.4 — Rated currents and the polarity test

Given. (a) Nameplate 10 kVA, 1100/110 V. (b) In a polarity test on a 220/110 V transformer, V_{12} = 220 V and V_{34} = 110 V. **Find** (a) rated currents; (b) what V_{13} readings of 330 V or 110 V mean. **Solution.** (a) 9.09 A and 90.9 A (Section 6). (b) 330 V = sum: terminals 1 and 4 have like polarity (additive). 110 V = difference: terminals 1 and 3 have like polarity (subtractive).

Example 17.5 — Perfectly coupled coils versus ideal transformer + magnetizing inductance

Given. $N_1 = 400$, $N_2 = 800$, $L_1 = 0.5$ H, $k = 1$ (so $L_2 = 2$ H, $M = 1$ H); $V_1 = 100$ V, 50 Hz; secondary load $R = 20\ \Omega$. **Find** I_1 and I_2 with the coupled-coil equations and with the "ideal transformer + L_1" model. **Solution.** Coupled coils: solve $\mathbf{V}_1 = j\omega L_1 \mathbf{I}_1 - j\omega M \mathbf{I}_2$, $0 = -j\omega M \mathbf{I}_1 + (j\omega L_2 + R)\mathbf{I}_2 \rightarrow \mathbf{I}_2 = 10.0\angle 0^\circ$ A, $\mathbf{I}_1 = 20.0 - j0.637$ A. Model: $a = 0.5$, $R' = a^2 R = 5\ \Omega$; $\mathbf{I}_1 = V_1/(j\omega L_1) + V_1/R' = -j0.637 + 20$ A; $\mathbf{I}_2 = a\,(V_1/R') = 10$ A. Identical (to 10^{-10}). **Interpretation.** The ideal transformer is not an approximation of perfectly coupled coils; it *is* perfectly coupled coils minus the magnetizing current.

Example 17.6 — Paralleling with wrong polarity (concept + estimate)

Given. Two identical 230/115 V units, each with series impedance $0.05 + j0.10\ \Omega$ referred to the 115 V side, are paralleled on the 230 V side but with one secondary reversed. **Find** the circulating current. **Solution.** Around the secondary loop the two 115 V EMFs now add: 230 V drives current through $2(0.05 + j0.10)\ \Omega$, $|Z| = 0.224\ \Omega$: $I_{cir} = 230/0.224 = 1029$ A — many times rated current (a 10 kVA unit's rated LV current is 87 A). Protection trips or the windings are damaged.

Example 17.7 — Ideal limit of the 2025 MT1 Q4(d) circuit

With the centre leg removed the two coils are perfectly coupled through a series core. In the ideal limit ($\mathcal{R}_s \to 0$): $I_2 = (N_1/N_2) I_1 = 0.5 \times 1.2 = 0.6$ A peak and $V_2 = 0.6 \times 20 = 12$ V peak. The finite core reluctance reduces this to 0.587 A and 11.7 V (Lesson 15, Example 15.4): the difference is the magnetizing current of Section 7.

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10. Optional practice

P17.1. An ideal 11 kV/415 V transformer supplies 200 kW at 0.8 pf lagging from its 415 V side at rated voltage. Find both currents and the kVA.

▼ **Solution**
$S = 200/0.8 = 250$ kVA. $I_{LV} = 250\,000/415 = 602$ A; $I_{HV} = 250\,000/11\,000 = 22.7$ A (ratio $a = 26.5$). Both sides carry 250 kVA at 0.8 pf.

P17.2. A 100 Ω load must look like 400 Ω to the source. Turns ratio? If instead it must look like 25 Ω?

▼ **Solution**
$a^2 = 400/100 \Rightarrow a = 2$ (source side has twice the turns); $a^2 = 0.25 \Rightarrow a = 0.5$.

P17.3 (concept, 2021 MT Q3(c)). Which equivalent-circuit elements disappear if there is no leakage, the core has infinite permeability and no hysteresis — and what remains?

▼ **Solution**
No leakage removes the leakage reactances; μ → ∞ removes the magnetizing reactance ($X_m \to \infty$, an open circuit); no hysteresis (and no eddy loss) removes the core-loss resistance ($R_c \to \infty$). What remains is the ideal transformer **plus the winding resistances** — the question does not assume zero resistance, so R_1 and R_2 stay.

P17.4. Why can a transformer not be used to step up a DC voltage? What happens if DC is applied to a transformer winding?

▼ **Solution**
$v = N\,d\Phi/dt$: constant v needs a linearly increasing flux, which saturates the core in milliseconds; after that the only limit to current is the small winding resistance — a large DC current flows and the secondary EMF falls to zero once the flux stops changing. DC is "transformed" only by switching it (choppers and switch-mode converters, Chapter V).

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11. Summary

Relation	Validity
$v_1/v_2 = N_1/N_2 = a$	Ideal: no leakage, no resistance; time-varying flux
$N_1 i_1 = N_2 i_2$, $i_1/i_2 = 1/a$	μ → ∞; i_1 into dot, i_2 out of dot
$S_1 = S_2$	Ideal (lossless, no stored energy)
$V_2' = aV_2$, $I_2' = I_2/a$, $Z_2' = a^2 Z_2$	Referring secondary → primary
Max power: $a^2 R_L = R_s$	Resistive source and load
$I_{rated} = S_{rated}/V_{rated}$ per winding	Nameplate

12. Vocabulary

English	Türkçe
transformer; ideal transformer	transformatör; ideal transformatör
primary / secondary winding	birincil (primer) / ikincil (sekonder) sargı
high-voltage / low-voltage winding	yüksek gerilim / alçak gerilim sargısı
turns ratio; transformation ratio	sarım oranı; dönüştürme oranı
polarity marking; dot	polarite işareti; nokta

English	Türkçe
referred quantity	yansıtılmış (indirgenmiş) büyüklük
impedance transformation / matching	empedans dönüşümü / uydurma
rating; rated current	anma değeri; anma akımı
core type / shell type	çekirdek tipi / mantel (kabuk) tipi
isolation	yalıtım (galvanik ayırma)

13. Sources

- LN7 viewer pp. 2–39 (slides 2–22): ideal transformer equations, current ratio and MMF balance, turns ratio, dots, referring rules, applications, assumptions. Notes.
- Sen §2 opening and §2.1 pp. 40–48 (viewer 64–72): construction, Eqs. 2.1–2.13, Example 2.1, polarity §2.1.2, parallel operation Fig. 2.9; ratings p. 51 (viewer 75).
- 2025 MT1 Q4(d) (Lesson 15 Ex. 15.4); 2021 MT Q3(c) p. 7.
- Verification: `lesson17_ideal_transformer.py`.

Lesson 18 — The practical transformer and its equivalent circuits

Chapter III: Transformers · Week 7 · Lesson 18 of 31

Assistant-written teaching note. Teacher source: LN8 (all 30 slides). Book: Sen §2.2 and §2.2.1 (printed pp. 48–51, viewer 72–75), Fig. 2.10–2.11. Exams: 2021 MT Q3(a)–(f); 2022 MT1 Q3 Part 1; MT I Fall 2018 Q2 Part I; MT I Fall 2019 Q2(f). Numbers: `lesson18_practical_transformer.py`.

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1. Place and goals

"Unfortunately ideal transformers do not exist" (LN7 slide 12). A real transformer's windings have resistance, some flux leaks, the core needs magnetizing current and dissipates core loss. LN8 slide 7 summarizes the remedy: **realistic equivalent circuit = ideal transformer + external impedances.** Each non-ideality gets one circuit element whose physical meaning you must be able to explain (MT 2018 Q2(a), 2021 MT Q3(a)).

After this lesson you should be able to:

1. Derive the exact (T) equivalent circuit, element by element, from the physical imperfections.
2. Refer the circuit to either side without mixing bases, and remove the ideal transformer.
3. Justify and use the approximate circuits (shunt branch at the terminals; series branch only; reactance only) and state when each is accurate.
4. Draw and interpret the phasor diagram of a loaded transformer.
5. Solve loaded-transformer problems for voltages, currents, losses and efficiency.

Prerequisites: Lesson 17; Lessons 13 (leakage), 15 (magnetizing current), 16 (core loss, R_c).

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2. Building the circuit one imperfection at a time

2.1 Winding resistance → R_1, R_2

Copper (or aluminium) windings have resistance, represented by lumped resistors in series with each winding (LN8 slide 8; Sen Fig. 2.10a). They cause the **copper loss** (bakır kaybı) $I_1^2R_1 + I_2^2R_2$, which grows with the square of the load current. In a step-down transformer, which winding has the larger resistance? The HV winding has more turns of thinner wire, so $R_1 > R_2$ — typically by about a^2, since a well-designed transformer roughly balances copper loss between the windings: $R_1 \approx a^2 R_2$ (Sen p. 53).

2.2 Leakage flux → X_1, X_2

Part of each winding's flux closes through air and links only that winding (Lesson 13; LN8 slide 9). Since the leakage path is mostly air, the leakage flux is proportional to current — a **constant leakage inductance** $L_{\ell 1} = N_1\Phi_{\ell 1}/i_1$, $L_{\ell 2} = N_2\Phi_{\ell 2}/i_2$ (Sen p. 49), i.e. leakage reactances $X_1 = \omega L_{\ell 1}$, $X_2 = \omega L_{\ell 2}$ in series with each winding. Leakage is minimized by winding the coils concentrically, close together (LN8 slides 10–11).

2.3 Finite core permeability → magnetizing reactance X_m

With μ finite, a net MMF — hence a primary current — is needed just to set up the mutual flux, even at no load (Lesson 15). That **magnetizing current** I_m lags the induced voltage by 90°, so it is represented by a shunt inductive reactance X_m across the induced voltage E_1 (LN8 slides 12, 19: "due to non-zero $\oint H\,dl$ in the core"). Lesson 17, Section 7 showed exactly this: perfectly coupled coils = ideal transformer + magnetizing inductance.

2.4 Core loss → core-loss resistance R_c

Hysteresis and eddy-current losses depend on the flux, i.e. on E_1 (Lesson 16). They are represented by a shunt resistance R_c across E_1, drawing the core-loss current I_c in phase with E_1 (LN8 slides 13–16, 19). Together:

$$\mathbf{I}_0 = \mathbf{I}_c + \mathbf{I}_m = \frac{\mathbf{E}_1}{R_c} + \frac{\mathbf{E}_1}{jX_m} \qquad \text{(exciting current, LN8 slide 17).}$$

2.5 The exact equivalent circuit

Putting the pieces together gives Figure 18.1 (LN8 slides 7, 18; Sen Fig. 2.10c): series $R_1 + jX_1$, then the shunt branch $R_c \parallel jX_m$ across E_1, then an ideal transformer $N_1 : N_2$, then series $R_2 + jX_2$ to the secondary terminals.

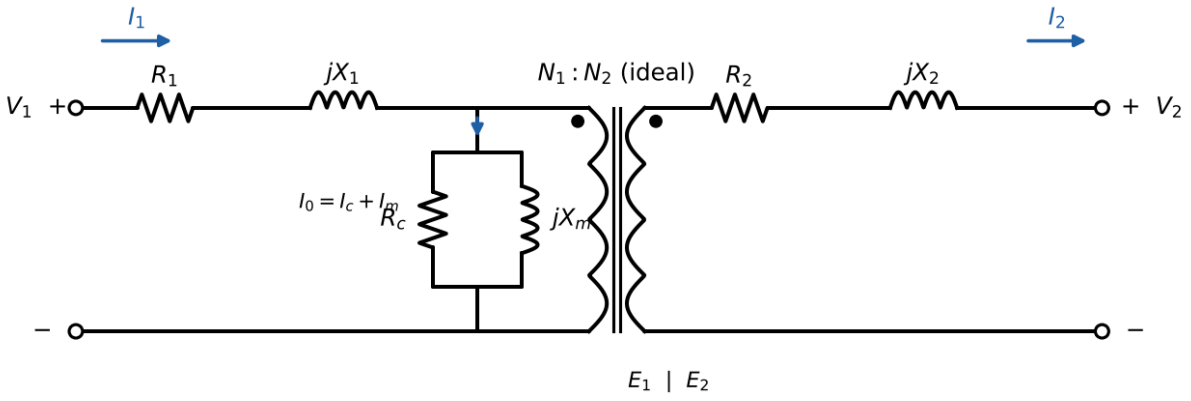


Figure 18.1. Exact (T) equivalent circuit: ideal transformer plus external impedances representing winding resistance (R_1, R_2), leakage (X_1, X_2), magnetization (X_m) and core loss (R_c).

Circuit equations (phasors, $\mathbf{I}_2$ leaving the secondary dot):

$$\mathbf{V}_1 = \mathbf{E}_1 + \mathbf{I}_1(R_1 + jX_1), \qquad \mathbf{E}_1 = a\mathbf{E}_2, \qquad \mathbf{E}_2 = \mathbf{V}_2 + \mathbf{I}_2(R_2 + jX_2),$$
$$\mathbf{I}_1 = \mathbf{I}_0 + \mathbf{I}_2/a. \tag{18.1}$$

Remember that E_1 and E_2 are the voltages induced by the *mutual* flux — they are not measurable at the terminals when current flows.

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3. Referring to one side

Following Lesson 17, move the ideal transformer to the secondary terminals by referring R_2, X_2 (and the load) to the primary with a^2 (LN8 slides 20–22, "multiply with $(N_p/N_s)^2$"; Sen Fig. 2.10d–e):

$$R_2' = a^2 R_2, \quad X_2' = a^2 X_2, \quad \mathbf{V}_2' = a\mathbf{V}_2, \quad \mathbf{I}_2' = \mathbf{I}_2/a.$$

The ideal transformer can then be dropped ("optional: transfer voltage and current to get rid of the ideal transformer", LN8 slide 22). Equally, everything can be referred to the secondary by dividing primary impedances by a^2. **Golden rule:** every element and every voltage/current in one equation must be referred to the *same* side.

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4. Approximate equivalent circuits

4.1 Shunt branch moved to the terminals ("cantilever" circuit)

Because $I_1(R_1 + jX_1)$ is only a few percent of V_1, $E_1 \approx V_1$. Moving the shunt branch to the input terminals changes the exciting current very slightly but makes the circuit much easier: the shunt branch sees V_1 directly, and the two series impedances can be combined (LN8 slides 23–26; Sen Fig. 2.11a):

$$R_{eq1} = R_1 + a^2 R_2, \qquad X_{eq1} = X_1 + a^2 X_2 \qquad \text{(LN8 slide 26)}. \tag{18.2}$$

Assumption (LN8 slide 24): "voltage drop in X_p and R_p is negligible" for the purpose of computing the exciting current. Example 18.1 quantifies the error: 0.18 % in $|V_1|$ for Sen's 10 kVA transformer at rated load.

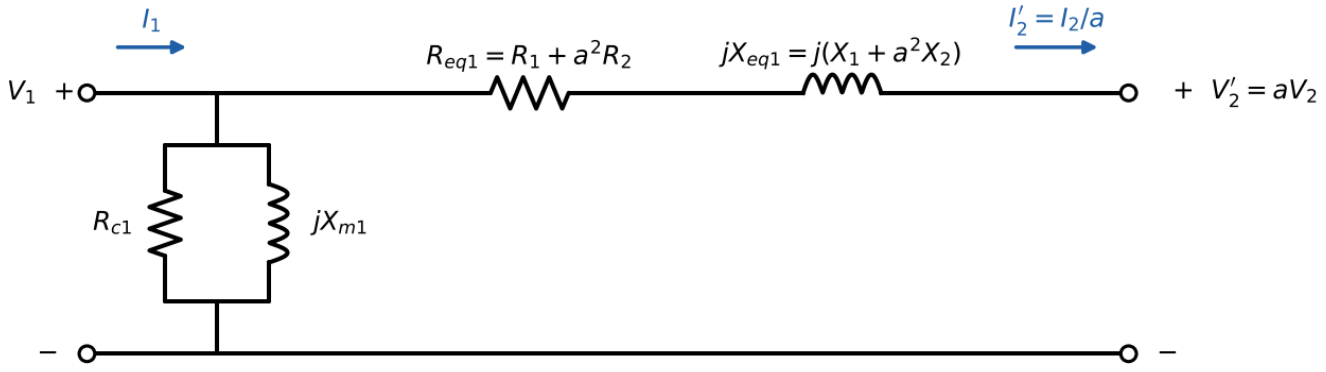


Figure 18.2. Approximate equivalent circuit referred to the primary, with the shunt branch at the input terminals and the series impedances combined.

4.2 Shunt branch neglected

If $I_0 \ll I_1$ (exciting current is typically 1–5 % of rated current; Sen: "less than 5 %"), drop the shunt branch for load-flow and regulation calculations (LN8 slide 27; Sen Fig. 2.11b–c): the transformer is just a series impedance $Z_{eq} = R_{eq} + jX_{eq}$. Keep the core loss separately when computing efficiency.

4.3 Reactance only

Large transformers (> 100 kVA) have efficiencies above 99 %, so $R_{eq} \ll X_{eq}$ and the resistance can be neglected for voltage and current calculations (LN8 slides 28–29). Not for efficiency, obviously.

Circuit	Use	Error source
Exact T	reference, any condition	—
Shunt at terminals	standard exam/design model; tests (Lesson 19)	exciting current computed with V_1 instead of E_1
Series only ($R_{eq} + jX_{eq}$)	voltage regulation, load flow, feeders	exciting current ignored (add P_{core} for efficiency)
jX_{eq} only	large units, fault and power-system studies	copper loss ignored

5. The phasor diagram of a loaded transformer

Draw it in the order the equations are solved (Figure 18.3, for Sen's 10 kVA transformer at rated current, 0.8 pf lagging):

1. Reference: secondary (referred) voltage $\mathbf{V}_2'$ on the real axis.
2. Load current $\mathbf{I}_2'$ lagging by θ_2 (36.9°).
3. $\mathbf{E}_1 = \mathbf{V}_2' + \mathbf{I}_2'(R_2' + jX_2')$: add the resistive drop parallel to $\mathbf{I}_2'$ and the reactive drop 90° ahead of it.
4. Exciting current $\mathbf{I}_0$: $\mathbf{I}_c$ in phase with $\mathbf{E}_1$, $\mathbf{I}_m$ lagging $\mathbf{E}_1$ by 90°.
5. $\mathbf{I}_1 = \mathbf{I}_2' + \mathbf{I}_0$.
6. $\mathbf{V}_1 = \mathbf{E}_1 + \mathbf{I}_1(R_1 + jX_1)$.

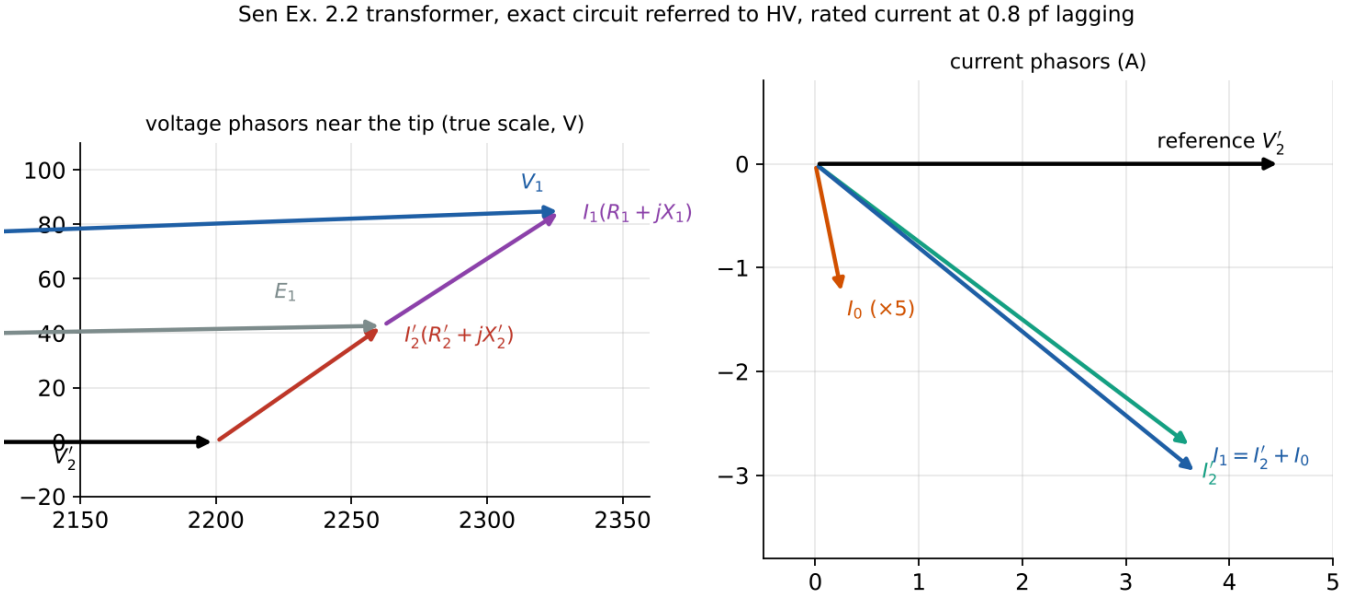


Figure 18.3. Left: tip region of the voltage phasors (true scale). Right: currents. V_1 is larger than V_2' and leads it; the exciting current is small and mostly magnetizing.

For a lagging load, the reactive drop adds almost in phase with $\mathbf{V}_2'$, so $|V_1| > |V_2'|$ markedly (positive regulation). For a leading load the reactive drop partly cancels and $|V_1|$ can be smaller than $|V_2'|$ (negative regulation) — Lesson 20.

6. Methods and common mistakes

Recipe for a loaded transformer problem.

1. Decide the side to work on (usually the side where the voltage is given).
2. Refer all impedances, the load and the known voltage to that side.
3. Use the approximate circuit unless told otherwise; state it.
4. Compute the series current from the load (if V_2 known) or from the source (if V_1 known), then voltages, then the exciting current from the terminal voltage.
5. Losses: copper I^2R_{eq}, core V^2/R_c (with V the voltage across the shunt branch in your chosen model). Efficiency $= P_{out}/(P_{out}+P_{cu}+P_{core})$.
6. Convert results to actual values; check with power balance at the terminals.

Mistakes: combining R_1 with R_2 (instead of R_2'); computing core loss with the load voltage when the shunt branch is placed at the primary; treating E_1 as a terminal voltage; forgetting that the phase of the load current is relative to *its own* voltage.

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7. Worked examples

Example 18.1 — Exact versus approximate circuit (Sen Example 2.2 transformer)

Given. 10 kVA, 2200/220 V transformer with HV-referred parameters $R_{eq}=10.4\ \Omega$, $X_{eq}=31.3\ \Omega$ (split equally, $R_1=R_2'=5.2$, $X_1=X_2'=15.65\ \Omega$), $R_c=48.4$ kΩ, $X_m=8.94$ kΩ (Lesson 19 derives these). It supplies rated current at 220 V, 0.8 pf lagging. **Find** V_1, I_1 and efficiency with the exact and the approximate circuit.

Solution (HV side). $\mathbf{V}_2'=2200\angle 0^\circ$ V, $\mathbf{I}_2'=4.545\angle -36.87^\circ$ A. *Exact:* $\mathbf{E}_1=2200+4.545\angle -36.87^\circ(5.2+j15.65)=2261.6+j42.7=2262.0\angle 1.08^\circ$ V. $\mathbf{I}_0=\mathbf{E}_1(1/R_c+1/jX_m)=0.257\angle -78.5^\circ$ A. $\mathbf{I}_1=4.741\angle -38.9^\circ$ A. $\mathbf{V}_1=\mathbf{E}_1+\mathbf{I}_1(5.2+j15.65)=2328.9\angle 2.09^\circ$ V. *Approximate:* $\mathbf{V}_1=2200+\mathbf{I}_2'(10.4+j31.3)=2324.8\angle 2.11^\circ$ V; $\mathbf{I}_1=\mathbf{I}_2'+\mathbf{V}_1/Z_{sh}=4.749\angle -38.9^\circ$ A. Efficiency: exact 96.04 %, approximate 96.08 %. **Check.** Exact P_{in} equals $P_{out}+I_1^2R_1+I_2'^2R_2'+E_1^2/R_c$ (to 10^{-9}). Referring the approximate solution to the LV side reproduces the actual secondary current $a\mathbf{I}_2'$.
Interpretation. The approximate circuit changes $|V_1|$ by 0.18 % and efficiency by 0.04 percentage points. The exciting current is 5.7 % of rated current here — on the high side for real transformers, which is why the difference shows at all.

Example 18.2 — A 220/110 V transformer with a load (2021 MT Q3(e,f))

Given. (exams/archive/MT/2021_Midterm_Q2_solution.pdf p. 8.) Approximate circuit referred to the 220 V side: $R_{eq}=2.4\ \Omega$, $X_{eq}=5.2\ \Omega$, $R_c=25$ kΩ, $X_m=50$ kΩ. Primary at 220 V, 50 Hz. (e) A 75 Ω, 0.8 pf lagging load (actual value on the 110 V side): find v_2 (magnitude and phase relative to v_1), total losses and efficiency. (f) Load removed: v_2 and losses; is efficiency meaningful?

Solution. $a=2$; $Z_L'=4(75\angle 36.87^\circ)=240+j180\ \Omega$.

e. $\mathbf{I}_2'=220/(242.4+j185.2)=0.7212\angle -37.38^\circ$ A; $\mathbf{V}_2'=\mathbf{I}_2'Z_L'=216.36\angle -0.51^\circ$ V; actual $\mathbf{V}_2=\mathbf{V}_2'/2=108.2\angle -0.51^\circ$ V. Copper loss $0.7212^2(2.4)=1.25$ W; core loss $220^2/25\,000=1.94$ W; $P_{out}=0.7212^2(240)=124.8$ W; $\eta=124.8/(124.8+3.18)=97.5$ %.
f. No load: $\mathbf{V}_2=110\angle 0^\circ$ V (no series drop in this model); losses = core loss 1.94 W; efficiency is 0 % (no output) — it is not a useful figure here; the loss itself is what matters (a standby loss of a plugged-in transformer). **Check.** $\mathrm{Re}\{V_1\mathbf{I}_1^*\}$ at the terminals equals output + losses exactly.

Example 18.3 — Magnetizing reactance from the core design (MT I Fall 2018 Q2, extended)

Given. (exams/archive/MT/MT I - Fall2018.pdf p. 2.) Primary 230 turns, secondary 120 turns, $V_1 = 230$ V, 50 Hz, peak mutual flux density 1.6 T, mean core length 1 m. Assume (our addition) $\mu_r = 4000$. **Find** (b) E_1; (c) core area and volume; and the magnetizing reactance and current.

Solution. (b) With the shunt branch at the terminals and no load, $E_1 \approx V_1 = 230$ V. (c) $A_c = E_1/(4.44 f N_1 B_{max}) = 230/(4.44 \times 50 \times 230 \times 1.6) = 2.81 \times 10^{-3}$ m² (28.1 cm²); $V_c = A_c l_c = 2.81 \times 10^{-3}$ m³. Magnetizing inductance $L_m = N_1^2 \mu A_c / l_c = 230^2 (4000\mu_0)(2.81 \times 10^{-3})/1 = 0.748$ H; $X_m = 235\ \Omega$; $I_m = 230/235 = 0.98$ A. **Interpretation.** The core size follows from voltage, frequency, turns and allowed flux density alone (Lesson 15). X_m then depends on μ_r: a gap or a poorer steel would lower X_m and raise I_m (Lesson 13, P13.4). Parts (d)–(f) of this question are solved in Lesson 20 (P20.4).

Example 18.4 — Reading the circuit physically (2022 MT1 Q3 Part 1; MT 2019 Q2(f))

(a) *For $N_1 \gg N_2$, compare X_1 and X_2.* Leakage inductance scales with the square of the turns for similar geometry, so $X_1 \gg X_2$ in ohms; referred to the same side they are comparable ($X_1 \approx a^2 X_2$). **(b)** *When is $v_1/v_2 = N_1/N_2$ exactly?* When there is no series voltage drop: zero winding resistance *and* zero leakage (or zero current, i.e. no load). Saying "if we ignore R_c" is not enough: R_c and X_m are shunt elements and do not affect the voltage ratio in the approximate model. **(c)** *A core material with a wider B–H loop at the same 1.5 T is used.* The hysteresis loss per cycle (loop area) increases, so the core loss rises and $R_c = E^2/P_c$ **decreases**. If the new loop also needs more H for 1.5 T, X_m decreases too. **(MT 2019 Q2(f))** *Which components are ignored if a coil is modelled with a lossless, leakage-free core?* $R_c \to \infty$ (no core loss) and $X_1, X_2 \to 0$ (no leakage); winding resistance only if stated.

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8. Optional practice

P18.1. A 50 kVA, 2400/240 V transformer has $R_1 = 0.75\ \Omega$, $X_1 = 1.0\ \Omega$, $R_2 = 0.0075\ \Omega$, $X_2 = 0.01\ \Omega$. Find R_{eq} and X_{eq} referred to each side.

▼ **Solution**
$a = 10$. HV side: $R_{eq1} = 0.75 + 100(0.0075) = 1.5\ \Omega$; $X_{eq1} = 1.0 + 100(0.01) = 2.0\ \Omega$. LV side: divide by 100: 0.015 Ω and 0.020 Ω. (Copper loss is equally split, as in a well-designed unit.)

P18.2. For the transformer of Example 18.1, estimate the error in I_0 caused by moving the shunt branch to the terminals at rated 0.8 pf lagging load.

▼ **Solution**
Exact $|I_0| = |E_1|/|Z_{sh}|$ with $|E_1| = 2262.0$ V; approximate uses $|V_1| = 2324.8$ V. The ratio is $2324.8/2262.0 = 1.028$: the approximate circuit overestimates the exciting current (and core loss) by about 2.8 % (core loss by 5.6 %, since it scales with V²). Small, because I_0 itself is small.

P18.3 (concept, MT 2018 Q2(a)). Give the physical meaning of each element of the exact circuit in terms of flux components and power-loss components.

▼ **Solution**

R_1, R_2: copper (ohmic) loss of the windings. X_1, X_2: leakage flux of each winding (no loss, stores energy in air). X_m: the mutual (magnetizing) flux in the core and the MMF needed to establish it (no loss). R_c: core loss (hysteresis + eddy currents) caused by the alternating mutual flux. Ideal transformer: the voltage/current transformation by the mutual flux.

9. Summary

Element	Represents	Typical size
R_1, R_2	winding resistance, copper loss ∝ I^2	R_{eq} ~ 0.5–2 % (pu)
X_1, X_2	leakage flux	X_{eq} ~ 2–10 % (pu)
X_m	magnetizing current (finite μ)	large, ~ 20–100 pu
R_c	core loss (hysteresis + eddy) ∝ V^2	very large, ~ 100–1000 pu
$R_{eq1} = R_1 + a^2 R_2$, $X_{eq1} = X_1 + a^2 X_2$	combined series branch	approximate circuit

10. Vocabulary

English	Türkçe
practical transformer	gerçek (pratik) transformatör
equivalent circuit (exact / approximate)	eşdeğer devre (tam / yaklaşık)
winding resistance; copper loss	sargı direnci; bakır kaybı
leakage reactance	kaçak reaktans
magnetizing reactance	mıknatıslanma reaktansı
core-loss resistance	nüve kaybı direnci (demir kaybı direnci)
exciting current	uyartım (boşta çalışma) akımı
shunt (parallel) branch	paralel (şönt) kol
series branch	seri kol
induced EMF	endüklenen EMK

11. Sources

- LN8 viewer pp. 2–54 (slides 2–30): losses review, laminations, non-idealities, equivalent circuit, winding resistance, leakage, magnetizing flux, core loss, excitation current, referring, simplifications 1–3, large transformers. Notes.
- Sen §2.2–2.2.1 pp. 48–51 (viewer 72–75), Figs. 2.10–2.11; leakage inductance definitions p. 49; $R_1 \approx a^2 R_2$ p. 53.
- 2021 MT Q3 (pp. 7–8); 2022 MT1 Q3 Part 1 (exams/archive/20221/Midterm-1 Q3 Solution.pdf p. 1); MT I Fall 2018 Q2 (p. 2); MT I Fall 2019 Q2(f) (p. 5).
- Verification: `lesson18_practical_transformer.py`.
- More solved archive problems: Ch II–III archive collection §4 (MT I Fall 2019 Q3, 1.25 MVA transformer).

Lesson 19 — Transformer tests: open-circuit and short-circuit

Chapter III: Transformers · Week 7 · Lesson 19 of 31

Assistant-written teaching note. Teacher source: LN9 (all 15 slides, including the 1000 VA example with answers). Book: Sen §2.2.2 (printed pp. 50–55, viewer 74–79), Example 2.2. Exams: 2022 Final Q1(a,b) (Jan 2022 paper); MT I Fall 2015 Q3(d,e); MT I Fall 2016 Q3(f); MT I Fall 2018 Q3(f); Jan 2023 Final Q1(c). Numbers: `lesson19_transformer_tests.py`.

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1. Place and goals

The equivalent circuit of Lesson 18 is only useful if we know its parameters. They could be computed from the design (Lesson 18, Example 18.3), but leakage reactance in particular is hard to calculate. LN9 asks: “how to test a closed system?” The answer mirrors the Thevenin/Norton method of circuit theory (LN9 slide 3): measure the transformer under two extreme conditions that each isolate one part of the circuit — **open circuit** (no load) and **short circuit** — with little power consumption.

After this lesson you should be able to:

1. Draw the wiring of both tests and state on which side and at what voltage/current each is performed, and why.
2. Explain which circuit elements each test “sees” and why the others can be neglected.
3. Extract R_c, X_m, R_{eq}, X_{eq} from voltmeter, ammeter and wattmeter readings, refer them to either side, and compute test power factors.
4. Interpret the results: exciting current percentage, per-unit impedance, consistency checks.
5. Run the logic backwards: predict test readings from known parameters, and extract parameters from other measurements.

Prerequisites: Lesson 18; Chapter I power and power-factor calculations.

---

2. Open-circuit (no-load) test

2.1 Setup and reasoning

One winding is left open; rated voltage at rated frequency is applied to the other (LN9 slides 5–8; Sen Fig. 2.12). The instruments — voltmeter V_{oc}, ammeter I_{oc}, wattmeter P_{oc} — are on the energized side. Usually the **LV side** is energized: rated LV voltage is easier and safer to supply, while the core flux is the same as with rated HV voltage on the HV side, because $\Phi_{max} = V/(4.44fN)$ and V/N is the same for both windings at rated voltage (Sen p. 52).

With the secondary open, the only current is the exciting current I_0 (a few percent of rated). In the exact circuit it flows through $R_1 + jX_1$ (negligible drop) into the shunt branch; in the approximate circuit with the shunt branch at the terminals, the series branch carries no current at all. So:

- The **wattmeter reads the core loss**: $P_{oc} = V_{oc}^2/R_c$ (the copper loss $I_0^2 R_1$ is negligible).
- The ammeter reads $I_0 = I_c + I_m$ (phasor sum).

2.2 Equations

$$R_c = \frac{V_{oc}^2}{P_{oc}}, \qquad I_c = \frac{V_{oc}}{R_c} = \frac{P_{oc}}{V_{oc}}, \qquad I_m = \sqrt{I_{oc}^2 - I_c^2}, \qquad X_m = \frac{V_{oc}}{I_m}. \tag{19.1}$$

Equivalent admittance form: $|Y| = I_{oc}/V_{oc}$, $G = P_{oc}/V_{oc}^2$, $B = \sqrt{|Y|^2 - G^2}$, $X_m = 1/B$. No-load power factor $\cos\theta_{oc} = P_{oc}/(V_{oc}I_{oc})$ is low (0.1–0.3): the exciting current is mainly magnetizing.

The parameters are **referred to the side where the test was made**. To refer them to the other side multiply or divide by a^2.

---

3. Short-circuit test

3.1 Setup and reasoning

One winding is short-circuited; a *reduced* voltage is applied to the other and raised until **rated current** flows (LN9 slides 9–10; Sen Fig. 2.13). LN9: “applied voltage is much smaller than the rated voltage, $V_1 \ll V_{rated}$” — typically 2–10 %. Applying rated voltage to a shorted transformer would drive 10–50 times rated current (LN9’s “transformer explosion” videos). The test is usually done from the **HV side**: rated HV current is the smaller current, and the required voltage (a few percent of HV) is conveniently large to measure.

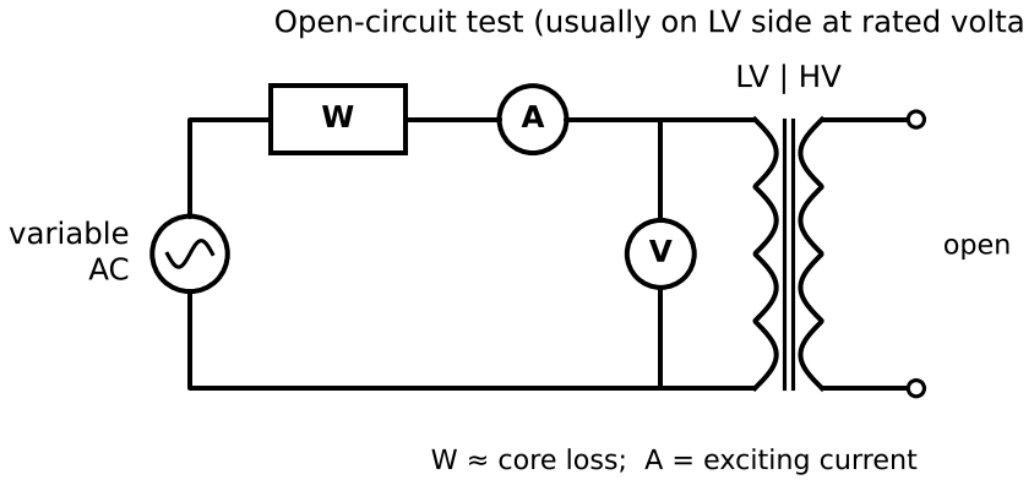
Because the applied voltage is a few percent of rated, the core flux is a few percent of normal, so:

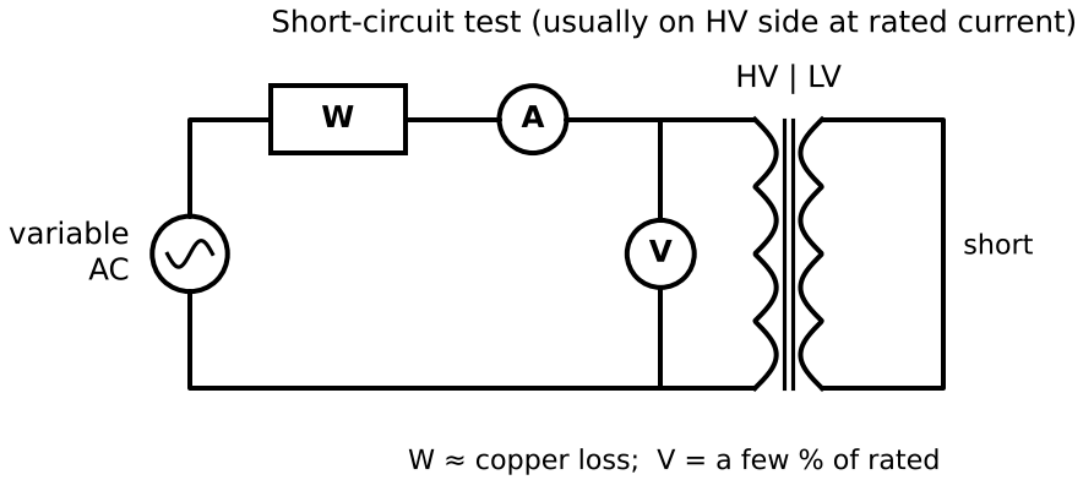
- the shunt-branch current is negligible (“neglect parallel branch! Current in the parallel branch is negligible due to small input voltage”, LN9 slide 10), and the core loss is negligible ($\propto$ V²);
- the **wattmeter reads the copper loss at rated current**: $P_{sc} = I_{sc}^2 R_{eq}$.

3.2 Equations

$$R_{eq} = \frac{P_{sc}}{I_{sc}^2}, \qquad |Z_{eq}| = \frac{V_{sc}}{I_{sc}}, \qquad X_{eq} = \sqrt{|Z_{eq}|^2 - R_{eq}^2}. \tag{19.2}$$

These are referred to the side where the test was made. The test gives only the *sums* $R_1 + a^2R_2$ and $X_1 + a^2X_2$; if the individual values are needed for the exact circuit, the usual assumption is $R_1 = a^2R_2$, $X_1 = a^2X_2$ (Sen p. 53) — or measure R_1, R_2 with DC and split only X_{eq}.





Figures 19.1–19.2. Test wiring (assistant redraw of LN9 slides 5 and 9 / Sen Figs. 2.12–2.13).

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4. Interpreting the results

- **Exciting current** I_{oc} as a percentage of the rated current of the tested winding (Sen Ex. 2.2(b): 5.5 %).
- **Percent (per-unit) impedance**: $Z_{pu} = |Z_{eq}|/Z_{base}$ with $Z_{base} = V_{rated}^2/S_{rated}$ of that side — equivalently V_{sc}/V_{rated} when the SC test is at rated current. It equals the per-unit voltage drop at rated current and sets the fault current: a short at the terminals of an infinite bus draws $1/Z_{pu}$ times rated current (Lesson 21).
- P_{oc} = core loss at rated voltage (constant loss); P_{sc} = full-load copper loss (variable loss, scales as load²). These two numbers alone give the efficiency curve and the load of maximum efficiency (Lesson 20).
- **Consistency checks:** $R_c \gg X_m \gg |Z_{eq}|$; both test power factors below 1; X_{eq} normally several times R_{eq} for larger units.

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5. Common mistakes

- Mixing sides: OC parameters come out on the OC side, SC parameters on the SC side. Refer them to one side before drawing the circuit.
- Using rated current in the OC test or rated voltage in the SC test.
- Computing $X_m = V/I_{oc}$ (that is $|Z_{shunt}|$, not X_m; subtract the core-loss current first).
- Forgetting that a SC test not done at rated current gives P_{sc} at *that* current: scale by $(I_{rated}/I_{sc})^2$ to get full-load copper loss.

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6. Worked examples

Example 19.1 — The LN9 example: 1000 VA, 230/115 V (answers on LN9 slide 14)

Given. (LN9 viewer pp. 20–23.) Both tests made “from the primary winding” (230 V side). No-load: $V_o = 230$ V, $I_o = 0.45$ A, $P_o = 30$ W. Short circuit: $V_s = 13.2$ V, $I_s = 6$ A, $P_s = 20.1$ W. **Find** the equivalent-circuit parameters.

Solution. *OC:* $R_c = 230^2/30 = 1763\ \Omega$; $I_c = 30/230 = 0.1304$ A; $I_m = \sqrt{0.45^2 - 0.1304^2} = 0.4307$ A; $X_m = 230/0.4307 = 534\ \Omega$. No-load pf $= 30/(230 \times 0.45) = 0.29$. *SC:* $R_{eq} = 20.1/6^2 = 0.558\ \Omega$; $|Z_{eq}| = 13.2/6 = 2.2\ \Omega$; $X_{eq} = \sqrt{2.2^2 - 0.558^2} = 2.13\ \Omega$. SC pf 0.25. All four values agree with LN9's answers (1763 Ω, j534 Ω, 0.558 Ω, j2.13 Ω), referred to the 230 V side. **Check.** The admittance method gives the same X_m. Referred to the 115 V side (÷4): $R_c = 441\ \Omega$, $X_m = 133.5\ \Omega$, $R_{eq} = 0.140\ \Omega$, $X_{eq} = 0.532\ \Omega$. **Remarks.** (i) The rated current of the 230 V winding is $1000/230 = 4.35$ A, so the SC test was made at 138 % of rated current — legitimate because the series branch is linear, but full-load copper loss is $4.35^2 \times 0.558 = 10.6$ W, not 20.1 W (used in Lesson 20). (ii) Here the OC test was made on the HV side at its rated voltage, which also gives rated flux. (iii) $|Z_{eq}|/Z_{base} = 2.2/52.9 = 4.2$ %.

Example 19.2 — Sen Example 2.2: 10 kVA, 2200/220 V

Given. OC (HV open, on LV): 220 V, 2.5 A, 100 W. SC (LV shorted, on HV): 150 V, 4.55 A, 215 W. **Find** (a) the approximate circuits referred to both sides; (b) exciting current as % of rated; (c) test power factors.

Solution. Rated currents: HV 4.55 A, LV 45.5 A; $a = 10$.

a. OC (LV): $R_{cL} = 220^2/100 = 484\ \Omega$; $I_c = 0.455$ A; $I_m = \sqrt{2.5^2 - 0.455^2} = 2.458$ A; $X_{mL} = 89.5\ \Omega$ (Sen rounds I_m to 2.46 A and gets 89.4 Ω). HV: $R_{cH} = 48.4$ kΩ, $X_{mH} = 8.95$ kΩ. SC (HV): $R_{eqH} = 215/4.55^2 = 10.39\ \Omega$; $|Z_{eqH}| = 150/4.55 = 32.97\ \Omega$; $X_{eqH} = 31.29\ \Omega$. LV: $R_{eqL} = 0.104\ \Omega$, $X_{eqL} = 0.313\ \Omega$.
b. $2.5/45.5 = 5.5$ %. (c) No-load pf $100/(220 \times 2.5) = 0.182$; SC pf $215/(150 \times 4.55) = 0.315$. **Interpretation.** The SC voltage is $150/2200 = 6.8$ % of rated: $Z_{pu} = 0.068$.

Example 19.3 — 10 kVA, 2500/250 V transformer (Jan 2022 Final Q1(a,b))

Given. (exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf p. 2.) OC: 250 V, 0.8 A, 50 W. SC: 80 V, 4 A, 80 W. **Find** (a) the side of each test and why; (b) parameters referred to the LV side.

Solution. (a) Rated currents: HV $10\,000/2500 = 4$ A, LV 40 A. The OC test reads 250 V = rated LV voltage, so it was done on the **LV side** (safe, convenient voltage, rated flux). The SC test current of 4 A is rated HV current at only 80 V (3.2 % of 2500 V), so it was done on the **HV side** with the LV side shorted (small current, measurable voltage).

b. OC → LV directly: $R_c = 250^2/50 = 1250\ \Omega$; $|Y| = 0.0032$ S, $G = 0.0008$ S, $B = 0.0031$ S ⇒ $X_m = 322.7\ \Omega$. SC → HV: $R_{eqH} = 80/16 = 5\ \Omega$, $|Z_{eqH}| = 20\ \Omega$, $X_{eqH} = 19.36\ \Omega$; referred to LV (÷100): $R_{eq} = 0.050\ \Omega$, $X_{eq} = 0.194\ \Omega$. **Interpretation.** Both test power factors are 0.25; $Z_{pu} = 20/625 = 3.2$ %. Parts (c)–(h) build a Δ–Y bank from three of these units (Lesson 21, Example 21.5).

Example 19.4 — Parameters from an OC test and a load test (MT I Fall 2015 Q3(d,e))

Given. (exams/archive/MT/MT I - Fall2015.pdf pp. 4–5.) 2500/500 V, 50 Hz. (d) OC test at rated voltage and frequency (shunt branch at the primary terminals): 800 W, pf 0.16 lagging. (e) With a resistive load: secondary 473 V, 50 A, unity pf; primary at rated voltage, leading the secondary voltage by 1.8°. **Find** (d) R_c, X_m; (e) the primary current (magnitude and phase) and the series impedance referred to the primary.

Solution. (d) Test on the primary at 2500 V: $I_0 = 800/(2500 \times 0.16) = 2.0$ A; $R_c = 2500^2/800 = 7812.5$ Ω; $I_m = 2.0\sin(\cos^{-1} 0.16) = 1.974$ A ⇒ $X_m = 1266$ Ω.

e. $a = 5$. Reference $\mathbf{V}_2' = 5(473) = 2365\angle 0^\circ$ V; $\mathbf{I}_2' = 50/5 = 10\angle 0^\circ$ A; $\mathbf{V}_1 = 2500\angle 1.8^\circ$ V. (i) $\mathbf{I}_1 = \mathbf{I}_2' + \mathbf{V}_1/R_c + \mathbf{V}_1/(jX_m) = 10.38 - j1.96 = 10.57\angle -10.7^\circ$ A. (ii) $Z_{eq1} = (\mathbf{V}_1 - \mathbf{V}_2')/\mathbf{I}_2' = (2498.8 + j78.5 - 2365)/10 = 13.38 + j7.85$ Ω. **Check.** $\mathrm{Re}\{\mathbf{V}_1\mathbf{I}_1^*\} = 23\,650 + 800 + 10^2(13.38)$ W ✓ (efficiency 91.7 %). **Interpretation.** A single load test with a phase measurement replaces the SC test: the 1.8° angle is exactly the information that separates R_{eq} from X_{eq}.

Example 19.5 — Predicting the readings of a short-circuit test (MT I Fall 2016 Q3(f))

Given. (exams/archive/MT/MT I - Fall2016.pdf p. 5.) 3600/240 V, 12 kVA; $Z_1 = 23.06 + j22.5$ Ω, $Z_2 = 0.1025 + j0.1$ Ω. The impedances came from a SC test made from the **LV** side with the HV side shorted. **Find** the instrument readings. **Solution.** $a = 15$. Referred to LV: $Z_{eq2} = Z_1/225 + Z_2 = 0.2050 + j0.2000$ Ω, $|Z_{eq2}| = 0.2864$ Ω. At rated LV current $12\,000/240 = 50$ A: ammeter 50 A, voltmeter $50 \times 0.2864 = 14.3$ V, wattmeter $50^2 \times 0.205 = 512$ W. **Interpretation.** Testing from the LV side needs 50 A at 14 V — feasible but awkward compared with 3.33 A at 215 V from the HV side; that is why the HV side is usual.

Example 19.6 — Why the shunt branch is ignored in the SC test (quantitative)

With the LN9 transformer, at the SC test voltage of 13.2 V the shunt branch would draw $|13.2/1763 + 13.2/j534| = 0.026$ A (0.4 % of the 6 A test current) and dissipate $13.2^2/1763 = 0.10$ W (0.5 % of the 20.1 W reading). Neglecting it is justified (Jan 2023 Final Q1(c) asks exactly this reasoning).

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7. Optional practice

P19.1. A 25 kVA, 2400/240 V transformer: OC on LV: 240 V, 3.2 A, 165 W; SC on HV: 55 V, 10.4 A, 360 W. Find all parameters referred to the HV side and the per-unit impedance.

▼ **Solution**
OC (LV): $R_c = 240^2/165 = 349$ Ω; $I_c = 0.6875$ A; $I_m = \sqrt{3.2^2 - 0.6875^2} = 3.125$ A; $X_m = 76.8$ Ω. HV (×100): $R_c = 34.9$ kΩ, $X_m = 7.68$ kΩ. SC (HV, rated current $25\,000/2400 = 10.42$ A ≈ 10.4 A): $R_{eq} = 360/10.4^2 = 3.33$ Ω; $|Z| = 55/10.4 = 5.29$ Ω; $X_{eq} = 4.11$ Ω. $Z_{base,HV} = 2400^2/25\,000 = 230.4$ Ω ⇒ $Z_{pu} = 0.023$ (2.3 %). *Values checked in the Lesson 19 script.*

P19.2. A SC test was made at 50 % of rated current and read 120 W. What is the full-load copper loss?

▼ **Solution**
Copper loss ∝ I²: $120 \times (1/0.5)^2 = 480$ W.

P19.3 (MT I Fall 2018 Q3(f)). You do not know the parameters of a 34.5 kV/230 V distribution transformer. Which tests, which instruments, and where?

▼ **Solution**
OC test on the 230 V side at rated voltage and frequency with voltmeter, ammeter and wattmeter on that side (34.5 kV side open — but dangerous: it carries 34.5 kV, insulate/guard it). SC test on the 34.5 kV side with the 230 V side shorted, raising a variable low voltage (a few percent of 34.5 kV, i.e. around 1–2 kV) until rated HV current flows; voltmeter, ammeter

and wattmeter on the HV side (via instrument transformers in practice). Draw as Figures 19.1–19.2.

P19.4 (concept). Why is the core loss measured in the OC test equal to the core loss at full load, but the copper loss measured in the SC test not equal to the copper loss at no load?

▼ **Solution**
Core loss depends on flux, set by voltage (Lesson 16); in normal operation the voltage stays near rated from no load to full load, so OC-test core loss applies at all loads (constant loss). Copper loss depends on current squared; the SC test at rated current gives the full-load value, which falls to almost zero at no load (variable loss).

8. Summary

Test	Side, condition	Reads	Gives
Open circuit	usually LV, rated V and f, other side open	V_{oc}, I_{oc} (exciting), P_{oc} (core loss)	$R_c = V^2/P$, $X_m = V/\sqrt{I^2 - (P/V)^2}$
Short circuit	usually HV, rated I, other side shorted, V ≈ 2–10 %	V_{sc}, I_{sc}, P_{sc} (copper loss)	$R_{eq} = P/I^2$, $X_{eq} = \sqrt{(V/I)^2 - R_{eq}^2}$

Parameters belong to the side tested; $Z_{pu} = V_{sc}/V_{rated}$ at rated current.

9. Vocabulary

English	Türkçe
open-circuit (no-load) test	açık devre (boşta çalışma) deneyi
short-circuit test	kısa devre deneyi
wattmeter, ammeter, voltmeter	wattmetre, ampermetre, voltmetre
equivalent circuit parameters	eşdeğer devre parametreleri
impedance voltage	kısa devre (empedans) gerilimi
per-unit impedance	birim değer (pu) empedans
no-load power factor	boşta güç faktörü

10. Sources

- LN9 viewer pp. 4–23 (slides 2–14): Thevenin analogy, test procedures, OC/SC equivalent circuits, 1000 VA example and answers. Notes.
- Sen §2.2.2 pp. 50–55 (viewer 74–79): parameter determination, ratings, OC and SC tests, Example 2.2.
- Jan 2022 Final Q1(a,b) p. 2; MT I Fall 2015 Q3(d,e) pp. 4–5; MT I Fall 2016 Q3(f) p. 5; MT I Fall 2018 Q3(f) p. 6; Jan 2023 Final Q1(c) (exams/archive/20221/EE361_Final_2022_Q4_Solution.pdf p. 3).
- Verification: `lesson19_transformer_tests.py`.

Lesson 20 — Transformer losses, efficiency and voltage regulation

Chapter III: Transformers · Week 8 · Lesson 20 of 31

Assistant-written teaching note. Teacher source: LN10 (all 36 slides; the regulation and efficiency examples use the LN9 1000 VA transformer). Book: Sen §2.3–2.4 (printed pp. 55–64, viewer 79–88), Examples 2.3–2.5. Exams: MT I Fall 2016 Q3; MT I Fall 2017 Q3; MT I Fall 2018 Q2 (Part III) and Q3; 2022 MT1 Q3 Part 2. Numbers: `lesson20_efficiency_regulation.py`*.*

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1. Place and goals

With the equivalent circuit (Lesson 18) and its parameters (Lesson 19) we can now predict how a transformer *performs*. LN10 slide 13 states the design goal in one line: **the best power transformer has efficiency as high as possible and voltage regulation as low as possible.** This lesson defines both figures of merit precisely, derives the conditions that govern them, and works through the examination problems that combine them.

After this lesson you should be able to:

1. Define voltage regulation, compute it exactly and approximately for lagging, unity and leading loads, and explain its sign with a phasor diagram.
2. Find the load power factor that maximizes regulation and the leading power factor that makes it zero.
3. Separate constant (core) and variable (copper) losses, compute efficiency at any load and power factor, and derive the maximum-efficiency conditions.
4. Compute all-day (energy) efficiency for a load cycle (Sen extension).
5. Solve multi-part exam problems with feeders, capacitors and measured waveforms.

Prerequisites: Lessons 18–19; Chapter I complex power and power-factor correction.

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2. Voltage regulation

2.1 Definition

As load current flows, the series impedance causes a voltage drop, so the secondary voltage changes between no load and full load (LN10 slide 2). The **voltage regulation** (gerilim regülasyonu) is

$$\mathrm{VR} = \frac{|V_{2,NL}| - |V_{2,FL}|}{|V_{2,FL}|} \times 100\% \qquad \text{(LN10 slide 3; Sen Eq. 2.16).} \tag{20.1}$$

Magnitudes matter, because loads care about the size of their voltage. Smaller is better; an ideal transformer has VR = 0. The load voltage is normally taken as **rated** voltage and the primary voltage is held at whatever value gives that; when the load is removed the secondary voltage becomes $|V_1|/a$. With the series-only (or shunt-at-terminals) model, referred to the primary:

$$\mathbf{V}_1 = \mathbf{V}_2' + \mathbf{I}_2'(R_{eq1} + jX_{eq1}), \qquad \mathrm{VR} = \frac{|\mathbf{V}_1| - |\mathbf{V}_2'|}{|\mathbf{V}_2'|} \qquad \text{(Sen Eqs. 2.19–2.21).} \tag{20.2}$$

The same number results on either side: $|V_2'| = a|V_2|$ and $|V_1|/a$ is the no-load secondary voltage.

2.2 Why the power factor matters

Figure 20.1 (LN10 slide 5) shows the phasors with $\mathbf{V}_2'$ as reference. The resistive drop IR_{eq} is parallel to the current; the reactive drop jIX_{eq} leads the current by 90°.

- **Lagging load:** $\mathbf{I}$ lags $\mathbf{V}_2'$; the reactive drop points largely *along* $\mathbf{V}_2'$ → $|V_1| \gg |V_2'|$ → large positive VR.
- **Unity pf:** the resistive drop adds in phase, the reactive drop at 90° adds little magnitude → small positive VR.
- **Leading load:** the reactive drop points partly *against* $\mathbf{V}_2'$ → $|V_1|$ can be smaller than $|V_2'|$ → **negative VR**: the voltage *rises* when the load is connected (LN10 slide 6: “capacitive: VR = ?”).

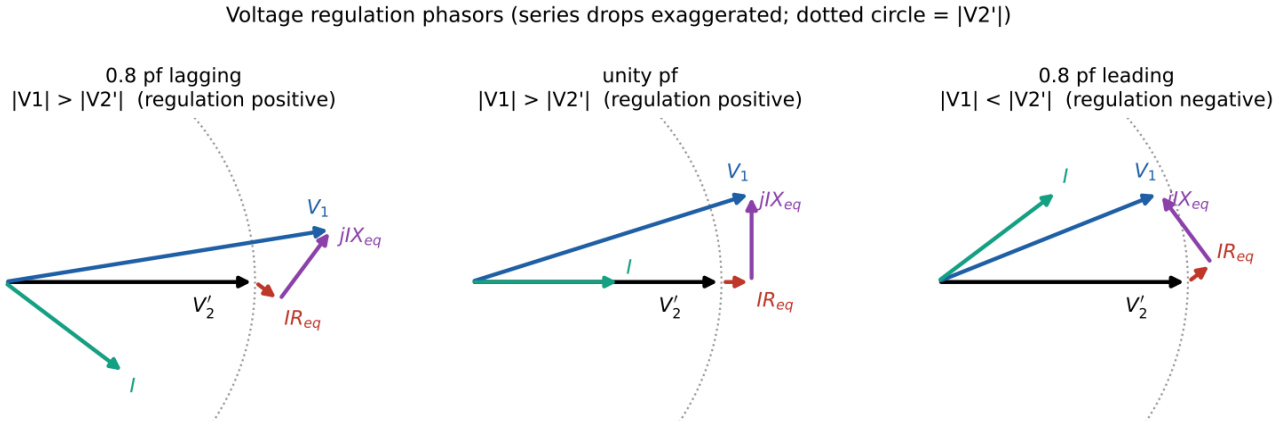


Figure 20.1. Phasor diagrams for lagging, unity and leading loads (drops exaggerated for visibility). The dotted circle has radius $|V_2'|$: V_1 outside the circle means positive regulation.

2.3 The approximate formula

Write $\mathbf{I} = I\angle -\theta$ (θ > 0 lagging). Expanding (20.2):

$$\mathbf{V}_1 = (V_2' + IR\cos\theta + IX\sin\theta) + j\,(IX\cos\theta - IR\sin\theta)\,.$$

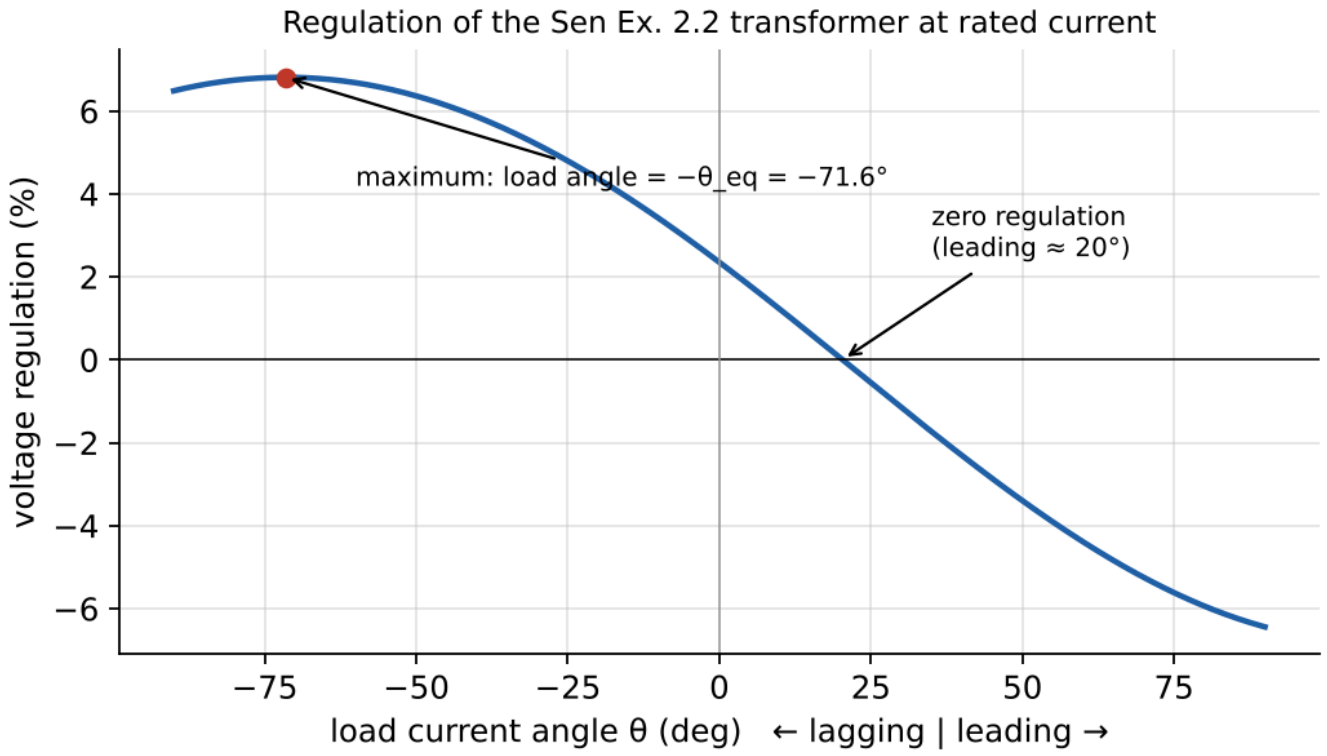
Since the imaginary part is small compared with V_2', $|\mathbf{V}_1| \approx V_2' + IR\cos\theta + IX\sin\theta$ and

$$\boxed{\mathrm{VR} \approx \frac{I(R_{eq}\cos\theta + X_{eq}\sin\theta)}{V_2'}} \qquad (\theta > 0 \text{ lagging},\ \theta < 0 \text{ leading}). \tag{20.3}$$

It is accurate to a small fraction of a percentage point for normal transformers (Example 20.1 compares). Use the exact phasor form when asked for “exact” or when drops are large.

2.4 Maximum and zero regulation

From the phasor picture, $|V_1|$ is largest when the drop $\mathbf{I}Z_{eq}$ is in phase with $\mathbf{V}_2'$, i.e. when the load angle equals minus the impedance angle: **maximum regulation occurs for a lagging load whose angle equals the transformer impedance angle** $\theta_{eq} = \tan^{-1}(X_{eq}/R_{eq})$ (Sen Eqs. 2.22–2.23). From (20.3), VR = 0 when $\tan\theta = -R_{eq}/X_{eq}$, i.e. at a *leading* angle of $\tan^{-1}(R_{eq}/X_{eq})$ (approximately). Figure 20.2 plots VR against load angle (LN10 slide 7 sketch).



LN10 slide 8 links this to the grid: a large PV plant raising feeder voltage, controlled by volt–var (reactive power) control — the same physics of current phase versus voltage drop.

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3. Losses and efficiency

3.1 Loss components

Loss	Cause	Depends on	Measured by
Core (iron) loss P_c	hysteresis + eddy currents (Lesson 16)	voltage and frequency (flux) — ≈ constant in service	OC test
Copper loss $P_{cu} = I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{eq1} = I_2^2 R_{eq2}$	winding resistance	current squared — “variable loss”	SC test (at rated current)

(LN10 slide 12: constant losses V^2/R_{core}; variable losses $I^2(R_1 + R_2')$; Sen Eqs. 2.27.) There are no rotational losses (no moving parts). Stray losses from leakage flux in tanks and clamps exist in large units but are lumped into the measured values here.

Temperature. Copper resistance rises with temperature: $R(T) = R(T_0)(1 + \alpha \Delta T)$ with $\alpha_{Cu} = 0.393$ %/°C, so resistivity rises about 31 % from 20 °C to 100 °C (LN10 slides 14–15). Copper losses are therefore quoted at a reference operating temperature. LN10 slides 16–29 explain why large transformers need oil, radiators and fans: heat generated scales with volume, heat dissipated with surface area (square–cube law), so bigger units cannot cool themselves passively.

3.2 Efficiency

$$\eta = \frac{P_{out}}{P_{in}} = \frac{P_{out}}{P_{out} + P_c + P_{cu}} = \frac{V_2 I_2 \cos\theta_2}{V_2 I_2 \cos\theta_2 + P_c + I_2^2 R_{eq2}} \tag{20.4}$$

(LN10 slides 10–11; Sen Eq. 2.29). For a fraction $x = I_2/I_{2,rated}$ of rated load with rated output voltage: $P_{out} = xS_{rated}\cos\theta_2$, $P_{cu} = x^2 P_{cu,FL}$, so

$$\eta(x) = \frac{xS\cos\theta_2}{xS\cos\theta_2 + P_c + x^2 P_{cu,FL}}. \tag{20.5}$$

The two test results (P_c from OC, $P_{cu,FL}$ from SC) are all you need. Well-designed large transformers exceed 99 % (LN8 slide 28).

3.3 Maximum efficiency (LN10 slides 33–35)

At what load is the efficiency maximum — full load, half load or no load? Hold V_2 and θ_2 fixed and differentiate (20.4) with respect to I_2. Writing $\eta = N/D$ with $N = V_2 I_2\cos\theta_2$, $D = N + P_c + I_2^2 R$:

$$\frac{d\eta}{dI_2} = \frac{(V_2\cos\theta_2)D - N(V_2\cos\theta_2 + 2I_2 R)}{D^2} = \frac{V_2\cos\theta_2\,\left(P_c - I_2^2 R\right)}{D^2} = 0$$

$$\Longrightarrow \quad \boxed{P_c = I_2^2 R_{eq2} \quad \text{(copper loss = core loss)}} \qquad x_{\eta_{max}} = \sqrt{\frac{P_c}{P_{cu,FL}}} \tag{20.6}$$

(Sen Eqs. 2.30–2.31c; the code confirms both conditions symbolically with SymPy). Intuition: losses per unit output, $(P_c + x^2 P_{cu,FL})/x$, are smallest when the falling term P_c/x and the rising term $xP_{cu,FL}$ are equal. At fixed current, η is also maximized by **unity power factor** (Sen Eq. 2.33: $d\eta/d\theta_2 = 0$ at θ₂ = 0). So the absolute maximum efficiency occurs at unity pf and the load where copper loss equals core loss (Figure 20.3).

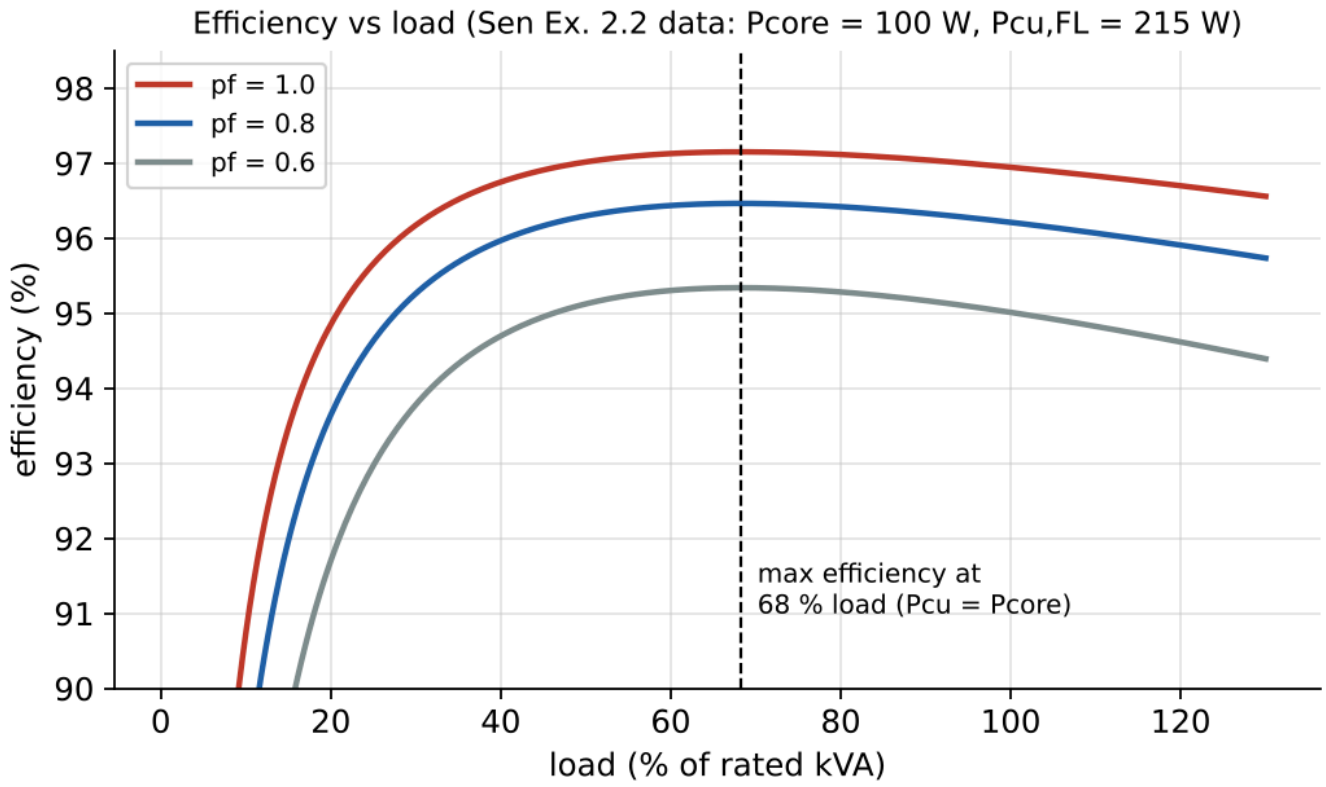


Figure 20.3. Efficiency versus load for Sen's 10 kVA transformer (LN10 slide 34's graph is missing in the supplied print; redrawn from data). The curve is flat around its maximum.

Design consequence: a **power transformer** working near full load all the time is designed for maximum efficiency near rated load; a **distribution transformer**, energized 24 h but lightly loaded much of the day, is designed with low core loss so that maximum efficiency occurs at its average load.

3.4 All-day (energy) efficiency (Sen §2.4.2 — extension)

$$\eta_{AD} = \frac{\text{energy out in 24 h}}{\text{energy out} + \text{core loss} \times 24\text{ h} + \sum x_k^2 P_{cu,FL} t_k} \tag{20.7}$$

Core loss counts for all 24 hours; copper loss only according to the load cycle (Example 20.5).

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4. Methods and common mistakes

Regulation: (1) work on one side; (2) set the load voltage at rated value as reference; (3) compute the load current phasor (angle *relative to its own voltage*); (4) $\mathbf{V}_1 = \mathbf{V}_2' + \mathbf{I}Z_{eq}$; (5) VR from magnitudes.

Efficiency: (1) P_{out} from load; (2) copper loss with the current through R_{eq}; (3) core loss with the voltage across the shunt branch in your model (Sen and LN10 use the rated-voltage OC value; MT 2016/2017 put the shunt at the primary, so use $|V_1'|^2/R_c$); (4) η.

Mistakes: using the leading/lagging sign the wrong way in (20.3); computing VR as a phasor difference instead of a magnitude difference; forgetting core loss in efficiency when the shunt branch was "neglected" for regulation; using P_{sc} measured at a current other than rated as full-load copper loss (Lesson 19 Example 19.1); claiming maximum efficiency at full load.

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5. Worked examples

Example 20.1 — Regulation of the LN10 transformer at three power factors (LN10 slide 9)

Given. The 1000 VA, 230/115 V transformer of LN9 (Lesson 19 Ex. 19.1: $R_{eq} = 0.558\ \Omega$, $X_{eq} = 2.128\ \Omega$ referred to 230 V). Output voltage adjusted to 115 V; rated current. **Find** VR for (a) 0.8 pf lagging; (b) unity; (c) 0.8 pf leading.

Solution. Work on the 115 V side: $R_{eq2} = 0.558/4 = 0.1396\ \Omega$, $X_{eq2} = 0.532\ \Omega$; $I = 1000/115 = 8.696$ A; reference $\mathbf{V}_2 = 115\angle 0^\circ$. $\mathbf{V}_1/a = 115 + 8.696\angle \mp 36.87^\circ (0.1396 + j0.532)$:

load	$\mathbf{V}_1/a$ (V)	VR exact	VR by (20.3)
0.8 lag	$118.78\angle 1.43^\circ$	**+3.29 %**	+3.26 %
unity	$116.31\angle 2.28^\circ$	**+1.14 %**	+1.06 %
0.8 lead	$113.28\angle 2.24^\circ$	**−1.49 %**	−1.57 %

(LN10 gives no printed answers; these are ours, verified in code.) **Interpretation.** The same current gives positive, small or negative regulation depending only on its phase.

Example 20.2 — Efficiency of the LN10 transformer (LN10 slide 32) and its maximum

Given. Same transformer, rated current, 0.8 pf lagging, output 115 V. Core loss 30 W (OC test at 230 V). **Find** η, and the load and value of maximum efficiency.

Solution. $P_{out} = 115 \times 8.696 \times 0.8 = 800$ W. Copper loss $= 8.696^2 \times 0.1396 = 10.55$ W (note: not the 20.1 W of the SC test, which was at 6 A on the 230 V side = 138 % of rated). $\eta = 800/(800 + 30 + 10.55) = 95.2$ %. *(Refinement: with the shunt branch at the primary, the primary voltage is* $2 \times 118.78 = 237.6$ *V and the core loss* $30(237.6/230)^2 = 32.0$ *W, giving 94.9 %.)* Maximum efficiency: $x = \sqrt{30/10.55} = 1.69$ — i.e. at 169 % of rated current, beyond rating. At rated load (x = 1) and unity pf the efficiency is still rising. **Interpretation.** This small transformer has relatively large core loss; its efficiency peak lies above rated load, so in practice it never reaches it. Large power transformers are designed with $x_{\eta max}$ near 0.5–1.

Example 20.3 — Sen Example 2.3: regulation at 75 % load, 0.6 pf lagging and leading

Given. Sen's 10 kVA, 2200/220 V transformer, $Z_{eqH} = 10.4 + j31.3\ \Omega$. **Find** VR at 75 % load, 0.6 pf (a) lagging, (b) leading. **Solution.** $I_H = 0.75 \times 4.55 = 3.41$ A. (a) $\mathbf{V}_H = 2200 + 3.41\angle -53.13^\circ(10.4 + j31.3) = 2307.0\angle 0.89^\circ$ V ⇒ VR = **4.86 %**. (b) $\mathbf{V}_H = 2200 + 3.41\angle 53.13^\circ(\ldots) = 2137.8\angle 2.48^\circ$ V ⇒ VR = **−2.83 %** (Sen: −2.82 %). **Extension.** At rated current the maximum VR is 6.82 %, at a lagging load angle of 71.6° = θ_{eq}; VR = 0 at about 20° leading (Figure 20.2).

Example 20.4 — Sen Example 2.4: efficiency and maximum efficiency

Given. Same transformer: $P_c = 100$ W, $R_{eq} = 10.4\ \Omega$ (HV) or $0.104\ \Omega$ (LV), $P_{cu,FL} = 215$ W. **Find** (a) η at 75 % load, 0.6 pf; (b) the output and value of maximum efficiency, and the load fraction. **Solution.** (a) $P_{out} = 0.75 \times 10\,000 \times 0.6 = 4500$ W; $P_{cu} = (0.75 \times 4.55)^2 \times 10.4 = 121$ W; $\eta = 4500/4721 = 95.32$ %. (b) Maximum at unity pf with $I_2^2(0.104) = 100$ ⇒ $I_2 = 31.0$ A; $P_{out} = 220 \times 31.0 = 6822$ W; $\eta_{max} = 6822/(6822 + 200) = 97.15$ % at **68.2 %** of full load. Check: $x = \sqrt{100/215} = 0.682$ ✓.

Example 20.5 — All-day efficiency (Sen Example 2.5, extension)

Given. 50 kVA, 2400/240 V; $P_c = 200$ W; $P_{cu,FL} = 500$ W. Load cycle: 0 % for 6 h; 50 % at pf 1 for 6 h; 75 % at 0.8 lag for 6 h; 100 % at 0.9 lag for 3 h; 110 % at pf 1 for 3 h. **Solution.** Energy out $= 0.5(50)(6) + 0.75(50)(0.8)(6) + 1(50)(0.9)(3) + 1.1(50)(1)(3) = 630$ kWh. Core loss $0.2 \times 24 = 4.8$ kWh. Copper loss $0.5\,[0.5^2(6) + 0.75^2(6) + 1^2(3) + 1.1^2(3)] = 5.7525$ kWh. $\eta_{AD} = 630/(630 + 10.55) = 98.35$ %.

Example 20.6 — Feeder, transformer, leading load (MT I Fall 2016 Q3)

Given. (exams/archive/MT/MT I - Fall2016.pdf p. 5.) 3600/240 V, 12 kVA, 50 Hz; $Z_1 = 23.06 + j22.5\ \Omega$, $Z_2 = 0.1025 + j0.1\ \Omega$, $R_c = 24\,633\ \Omega$ connected at the primary terminals; source feeder $Z_f = j31.5\ \Omega$ (HV side); load $\mathbf{V}_2 = 240\angle 0^\circ$ V, $\mathbf{I}_2 = 50\angle +60^\circ$ A. **Find** (a) the approximate circuit referred to the secondary; (b) the load impedance; (c) VR; (d) core loss, copper loss, η, comment; (e) source voltage and power factor; (f) SC-test readings (Lesson 19 Ex. 19.5).

Solution. $a = 15$. (a) $Z_{eq2} = Z_1/225 + Z_2 = 0.2050 + j0.2000\ \Omega$; $R_c' = 24\,633/225 = 109.48\ \Omega$ (at the referred primary terminals); $Z_f' = j0.14\ \Omega$.

b. $Z = 240/50\angle 60^\circ = 4.8\angle -60^\circ = 2.40 - j4.157\ \Omega$ (capacitive).
c. $\mathbf{V}_1' = 240 + 50\angle 60^\circ(0.205 + j0.2) = 236.87\angle 3.36^\circ$ V ⇒ VR = **−1.30 %** (leading load raises the voltage).

d. $P_{core} = 236.87^2/109.48 = 512.5$ W; $P_{cu} = 50^2(0.205) = 512.5$ W; $P_{out} = 240 \times 50\cos 60^\circ = 6000$ W; $\eta = 6000/7025 = 85.4$ %. Comment: copper loss equals core loss, so for *this current and power factor* the efficiency is at its maximum; but the poor load power factor (0.5) makes the output small for the current, so η is low. At unity pf and the same current, η would be $12\,000/13\,025 = 92.1$ %.
e. Source current (referred) $\mathbf{I}_s = \mathbf{I}_2 + \mathbf{V}_1'/R_c'$; $\mathbf{V}_s' = \mathbf{V}_1' + \mathbf{I}_s Z_f' \Rightarrow |V_s| = 3466$ V (actual, HV side), source pf 0.594 (leading). **Check.** Source complex power real part = 6000 + 512.5 + 512.5 W ✓.

Example 20.7 — Regulation, efficiency and a correcting capacitor (MT I Fall 2017 Q3)

Given. (exams/archive/MT/MT I - Fall2017.pdf pp. 5–7, with typed key.) 2200/220 V, 11 kVA; $R_1 = 2$, $X_1 = 10$, $R_2 = 0.02$, $X_2 = 0.1$ Ω; $R_c = 52.992$ kΩ at the primary terminals; X_m ignored. Load $Z = 4.4\angle 78.7^\circ$ Ω at $\mathbf{V}_2 = 220\angle 0^\circ$ V. **Find** (a) circuit referred to the secondary; (b) V_1'; (c) VR; (d) η; (e, bonus) capacitor for unity pf at the secondary, the new primary voltage and primary pf.

Solution. (a) $Z_{eq2} = (2 + j10)/100 + (0.02 + j0.1) = 0.04 + j0.2 = 0.204\angle 78.7^\circ$ Ω; $R_c' = 529.92$ Ω.

b. $\mathbf{I}_2 = 220/4.4\angle 78.7^\circ = 50\angle -78.7^\circ$ A; $\mathbf{V}_1' = 220 + 0.204\angle 78.7^\circ \times 50\angle -78.7^\circ = 220 + 10.2 = 230.2\angle 0^\circ$ V. (The load angle equals the impedance angle — Section 2.4's worst case: the drop is exactly in phase.)
c. VR = 10.2/220 = **4.64 %**.
d. $P_{out} = 220 \times 50\cos 78.7^\circ = 2155.4$ W; $P_{cu} = 50^2(0.04) = 100$ W; $P_c = 230.2^2/529.92 = 100$ W; $\eta = 2155.4/2355.4 = 91.5$ %. Comment: copper = core loss, so η is maximum for this pf, but the very low pf (0.196) wastes the transformer's current capacity.
e. Load reactive power $Q = 220 \times 50\sin 78.7^\circ = 10.79$ kvar ⇒ $C = Q/(V^2\omega) = 709$ μF (key: 708.9 μF). Now $I = P/V = 9.80$ A in phase with V_2: $\mathbf{V}_1' = 220 + 9.80(0.04 + j0.2) = 220.4\angle 0.51^\circ$ V, i.e. 2204 V actual; the primary current $\mathbf{I}_2 + \mathbf{V}_1'/R_c'$ is almost in phase: $\text{pf}_1 \approx 1.000$. Copper loss drops from 100 W to 3.8 W and η rises to 95.8 %. **Interpretation.** Power-factor correction at the load (Chapter I) also improves the transformer's regulation and efficiency, because the current through Z_{eq} falls fivefold.

Example 20.8 — METU campus distribution transformer (MT I Fall 2018 Q3)

Given. (exams/archive/MT/MT I - Fall2018.pdf pp. 4–6.) Single-phase model, 150:1 (34.5 kV/230 V), $R_1 = 9$ Ω, $R_2 = 0.0004$ Ω, $X_1 = 36$ Ω, $X_2 = 0.0016$ Ω, exciting branch neglected. **Find** (a) R_{eq}, X_{eq} referred to the MV side; (b) the secondary voltage when the transformer draws 30 A at unity pf from 34.5 kV; (c) the same at 30° lagging; (d) efficiencies; (e) yearly cost of losses at 30 A, 0.45 TL/kWh; (f) tests (Lesson 19 P19.3).

Solution. (a) $R_{eq1} = 9 + 150^2(0.0004) = 18$ Ω; $X_{eq1} = 36 + 22\,500(0.0016) = 72$ Ω. (b) Here the *primary* voltage is fixed: $\mathbf{V}_2' = 34\,500 - 30\angle 0^\circ(18 + j72) \Rightarrow |V_2| = 226.9$ V. (c) $\mathbf{I}_1 = 30\angle -30^\circ$: $|V_2| = 219.9$ V. Comment: the lagging current turns the large reactive drop toward the voltage, lowering the secondary voltage by a further 7 V. (d) Copper loss $30^2 \times 18 = 16.2$ kW in both cases; $P_{in} = 34\,500 \times 30 \times \cos\theta$: η = 98.4 % (unity), 98.2 % (lagging). (e) Energy lost per year $16.2\,\text{kW} \times 8760\,\text{h} = 141\,912$ kWh, cost ≈ 63 860 TL per transformer (core loss excluded as the exciting branch is neglected).

Example 20.9 — Efficiency from measured waveforms (2022 MT1 Q3 Part 2)

Given. (exams/archive/20221/Midterm-1 Q3 Solution.pdf pp. 2–3.) 1000 VA, 115:230 V, 50 Hz; DC-measured $R_1 = 0.25\ \Omega$, $R_2 = 1\ \Omega$. Measured: $v_1 = 171.25\cos(\omega t + 10.66^\circ)$, $i_1 = 11.7\cos(\omega t - 0.5^\circ)$, $v_2 = 325.30\cos\omega t$, $i_2 = 5.66\cos\omega t$. Approximate circuit with the shunt branch at the primary. **Find** (a) η; (b) copper and core losses; (c) R_c; (d) X_{eq}; (e) η if $X_{eq} = 0$ with the same secondary conditions.

Solution. RMS phasors: $V_1 = 121.1\angle 10.66^\circ$, $I_1 = 8.27\angle -0.5^\circ$, $V_2 = 230.0\angle 0^\circ$, $I_2 = 4.00\angle 0^\circ$.

a. $P_{in} = 121.1 \times 8.27\cos 11.16^\circ = 982.9$ W; $P_{out} = 230 \times 4.0 = 920.6$ W; $\eta = 93.7$ %.
b. $a = 0.5$; $R_{eq1} = 0.25 + 0.25(1) = 0.5\ \Omega$; $I_2' = I_2/a = 8.00$ A; $P_{cu} = 8.0^2 \times 0.5 = 32.0$ W; $P_{core} = 982.9 - 920.6 - 32.0 = 30.2$ W.
c. $R_c = 121.1^2/30.2 = 485\ \Omega$.
d. $Z_{eq1} = (\mathbf{V}_1 - a\mathbf{V}_2)/\mathbf{I}_2' = 0.499 + j2.80\ \Omega \Rightarrow X_{eq} \approx 2.8\ \Omega$ (the resistive part, 0.499 Ω, agrees with the DC measurement — a good consistency check).
e. With $X_{eq} = 0$: $V_1 = 115 + 8.0 \times 0.5 = 119.0$ V, core loss $30.2(119.0/121.1)^2 = 29.2$ W, copper loss unchanged: η = 93.8 %. **Why:** X_{eq} dissipates no power; it only changes the primary voltage needed, hence slightly the core loss through V_1^2/R_c. Its main effect is on regulation, not efficiency.

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6. Optional practice

P20.1. A 100 kVA transformer has $P_c = 600$ W and $P_{cu,FL} = 1500$ W. Find η at full load and half load, unity pf, and the maximum efficiency.

▼ **Solution**
Full load: $100/(100 + 0.6 + 1.5) = 97.94$ %. Half: $50/(50 + 0.6 + 0.375) = 98.09$ %. Maximum at $x = \sqrt{600/1500} = 0.632$: $63.2/(63.2 + 1.2) = 98.14$ %.

P20.2. Using (20.3), estimate VR at rated current for a transformer with $R_{eq} = 0.01$ pu and $X_{eq} = 0.05$ pu at 0.8 lag, unity and 0.8 lead; compare with the exact phasor result.

▼ **Solution**
VR ≈ $R\cos\theta + X\sin\theta$ in pu: $0.01(0.8) + 0.05(0.6) = 0.038$ (3.8 %); 0.01 (1 %); $0.008 - 0.03 = -0.022$ (−2.2 %). Exact phasor values: 3.86 %, 1.12 %, −2.09 %; the difference is the second-order term ≈ $\frac{1}{2}(X\cos\theta - R\sin\theta)^2$, which is always positive.

P20.3. At what leading power factor is the regulation of the Sen Ex. 2.2 transformer zero at rated current?

▼ **Solution**
Approximately $\tan\theta = R/X = 10.4/31.3 \Rightarrow \theta = 18.4°$ leading (pf 0.949); the exact phasor calculation gives 20.3° (pf 0.938), because the second-order term must be cancelled too (Figure 20.2).

P20.4 (MT I Fall 2018 Q2 Part III). The 230 V transformer of Lesson 18 Ex. 18.3 must work from 10 Hz to 50 Hz at 230 V without exceeding 1.6 T. Find the new core area and the flux density at 50 Hz; compare volumes.

▼ **Solution**

$B \propto V/(fNA)$, so the worst case is 10 Hz: $A = 230/(4.44 \times 10 \times 230 \times 1.6) = 0.0141$ m² — five times the 50 Hz design (2.81×10^{-3} m²); with l_c = 1 m the volume rises from 2.8 L to 14 L. At 50 Hz this core runs at only 1.6/5 = 0.32 T. Low-frequency operation at full voltage costs core material; that is why variable-frequency supplies reduce V with f.

7. Summary

Relation	Validity
VR = $(\lvert V_{2,NL}\rvert - \lvert V_{2,FL}\rvert)/\lvert V_{2,FL}\rvert$	definition (magnitudes)
$\mathbf{V}_1 = \mathbf{V}_2' + \mathbf{I}_2' Z_{eq1}$	series model
VR $\approx I(R\cos\theta + X\sin\theta)/V$	small drops; θ > 0 lagging
max VR at θ = θ_eq (lag); VR = 0 at leading θ ≈ $\tan^{-1}$(R/X)	series model
η = $P_{out}/(P_{out} + P_c + x^2 P_{cu,FL})$	rated voltage
η_max: $P_{cu} = P_c$, $x = \sqrt{P_c/P_{cu,FL}}$, unity pf	constant V
η_{AD} = energy out/(energy out + 24 h core + Σ copper)	load cycle (extension)

8. Vocabulary

English	Türkçe
voltage regulation	gerilim regülasyonu
efficiency; maximum efficiency	verim; maksimum verim
constant (core) loss / variable (copper) loss	sabit (demir) kayıp / değişken (bakır) kayıp
all-day efficiency	günlük (enerji) verimi
load cycle	yük çevrimi
power transformer / distribution transformer	güç transformatörü / dağıtım transformatörü
temperature coefficient	sıcaklık katsayısı
cooling (radiator, oil, fan)	soğutma (radyatör, yağ, fan)

9. Sources

- LN10 viewer pp. 5–51 (slides 2–36): regulation definition and phasors, sign with pf, PV/volt–var context, 1000 VA examples, efficiency, constant/variable losses, temperature, cooling/square–cube law, maximum efficiency. Notes.
- Sen §2.3–2.4 pp. 55–64 (viewer 79–88): Eqs. 2.14–2.34, Examples 2.3–2.5, Fig. 2.15.
- MT I Fall 2016 Q3 p. 5 (no key supplied); MT I Fall 2017 Q3 pp. 5–7 (typed key); MT I Fall 2018 Q2 p. 2 and Q3 pp. 4–6; 2022 MT1 Q3 Part 2 pp. 2–3 (handwritten key, partly illegible; values recomputed).
- Verification: `lesson20_efficiency_regulation.py` (includes SymPy derivation of the maximum-efficiency conditions).

- More solved archive problems: Ch II–III archive collection §4–7 (MT I Fall 2019 Q3, Final Fall 2016 Q1, Final Fall 2017 Q2 Part I, recitation regulation examples).

Lesson 21 — Three-phase transformers

Chapter III: Transformers · Week 9 · Lesson 21 of 31

Assistant-written teaching note. Teacher source: LN11 (all 29 slides, including the 3 × 2 MVA Δ–Y example, Sen Ex. 2.8 and a handwritten single-line example). Book: Sen §2.6–2.8 (printed pp. 66–84, viewer 90–108): connections, phase shift, single-phase equivalent, Examples 2.7–2.8, open-delta, three-phase core, harmonics, per-unit. Exams: Final Fall 2014 Q1; Final Fall 2017 Q1; Jan 2022 Final Q1(c)–(h); Jan 2023 Final Q1. Chapter I prerequisites: Lessons 7–9 (balanced three-phase, Y/Δ, per-phase analysis). Numbers: `lesson21_three_phase_transformers.py`.

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1. Place and goals

Bulk power is generated, transmitted and distributed in three-phase form (LN11 slide 2: more power per unit wire length, constant total power, rotating MMF for machines). Every voltage change in that system needs a **three-phase transformer**. Fortunately, nothing new is needed in the physics: a three-phase transformer is three single-phase transformers (or three phases on one core) whose windings are connected in Y or Δ. What is new is bookkeeping: line versus phase quantities, line-to-line versus turns ratios, and phase shifts.

After this lesson you should be able to:

1. Describe three-phase banks and three-limb cores and their trade-offs.
2. For Y–Y, Δ–Δ, Y–Δ and Δ–Y connections, compute winding and line voltages and currents, the turns ratio and the line-to-line ratio, and explain the 30° phase shift.
3. Build the per-phase (equivalent-Y) circuit referred to either side, including feeders and sources, and solve for voltages, regulation, losses and efficiency.
4. Use per-unit values for transformer impedance and fault current (the archive finals do; labelled as a book extension).
5. State the practical uses and drawbacks of each connection (harmonics, neutral, open-delta).

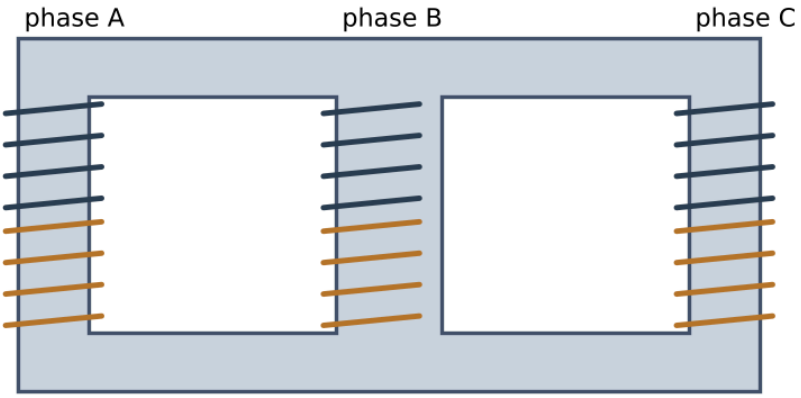
Prerequisites: Lessons 17–20; Chapter I Lessons 7–9.

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2. Construction

Bank of three single-phase units (Sen §2.6.1): three identical transformers, primaries and secondaries each connected in Y or Δ. If one unit fails it can be replaced alone, and with Δ–Δ the bank can even run on two units (open-delta).

Three-phase unit on one core (Sen §2.6.2; LN11 slides 8–11): start with three single-phase cores whose return legs carry Φ_a, Φ_b, Φ_c. For balanced sinusoidal voltages these fluxes sum to zero, so the common return leg can be removed; squeezing the middle unit between the others gives the familiar **three-limb core**, each limb carrying the HV and LV windings of one phase (Figure 21.1). It is “lighter, cheaper, more efficient” (LN11 slide 8) and takes less space; its drawback is that a fault in one phase takes the whole unit out of service. The outer limbs’ longer paths cause slightly unequal magnetizing currents (insignificant).



each limb carries the HV (dark) and LV (brown) windings of one phase; Φa+Φb+Φc = 0 so no return limb is needed

3. The four connections

3.1 Winding versus line quantities

From Chapter I: in **Y**, $|V_{LL}| = \sqrt{3}|V_{ph}|$ and $I_{line} = I_{ph}$; in **Δ**, $V_{LL} = V_{ph}$ and $|I_{line}| = \sqrt{3}|I_{ph}|$ (LN11 slides 4–5). Here "phase" means **the winding** of one single-phase unit. Each winding carries one third of the total power regardless of connection (LN11 slide 6: $S = 3V_{ph}I_{ph} = \sqrt{3}V_{LL}I_{line}$).

3.2 Turns ratio versus line-to-line ratio (LN11 slide 18)

Let each unit have turns ratio $a = N_1/N_2 = V_{1,ph}/V_{2,ph}$ (winding voltages). The line-to-line ratio of the bank depends on the connection:

Connection	$V_{1,LL}$	$V_{2,LL}$	**LL ratio** $V_{1,LL}/V_{2,LL}$	**Phase shift of LL voltages**	**Typical use (LN11 sl. 14–17, Sen)**
Y–Y	$\sqrt{3}V_{1,ph}$	$\sqrt{3}V_{2,ph}$	a	0°	rare: 3rd-harmonic problems unless a Δ tertiary is added
Δ–Δ	$V_{1,ph}$	$V_{2,ph}$	a	0°	medium voltage; can run open-delta at 57.7 %
Y–Δ	$\sqrt{3}V_{1,ph}$	$V_{2,ph}$	$\sqrt{3}\,a$	30°	step-down (HV neutral can be grounded)
Δ–Y	$V_{1,ph}$	$\sqrt{3}V_{2,ph}$	$a/\sqrt{3}$	30° (sign by convention)	step-up at generators; distribution (LV neutral for single-phase loads, e.g. Dyn11)

For Y–Δ and Δ–Y **the line-to-line voltage ratio is not the turns ratio**; the LL ratio is what matters for the system and is the ratio of the equivalent Y–Y transformer used in per-phase analysis (Sen Eqs. 2.39–2.40). Example 21.1 tabulates all four for the same single-phase units.

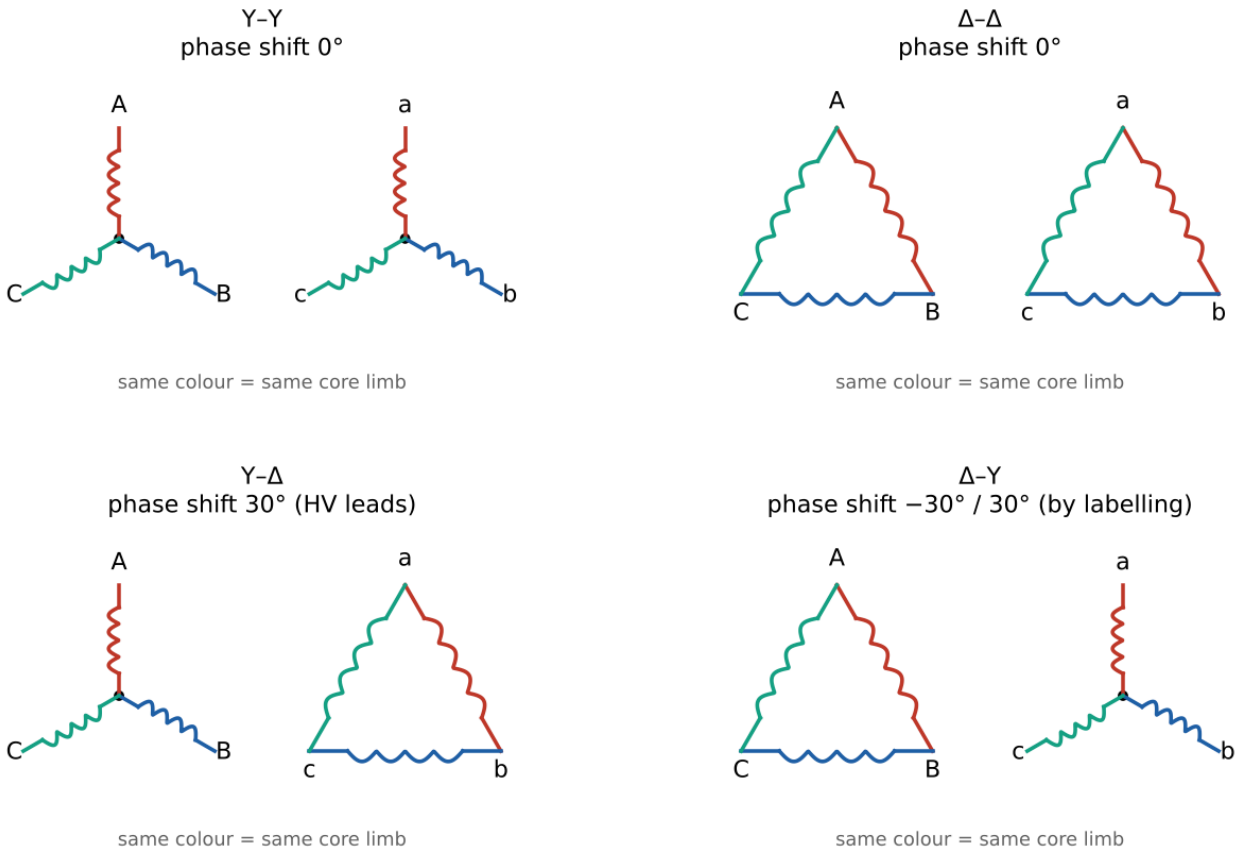
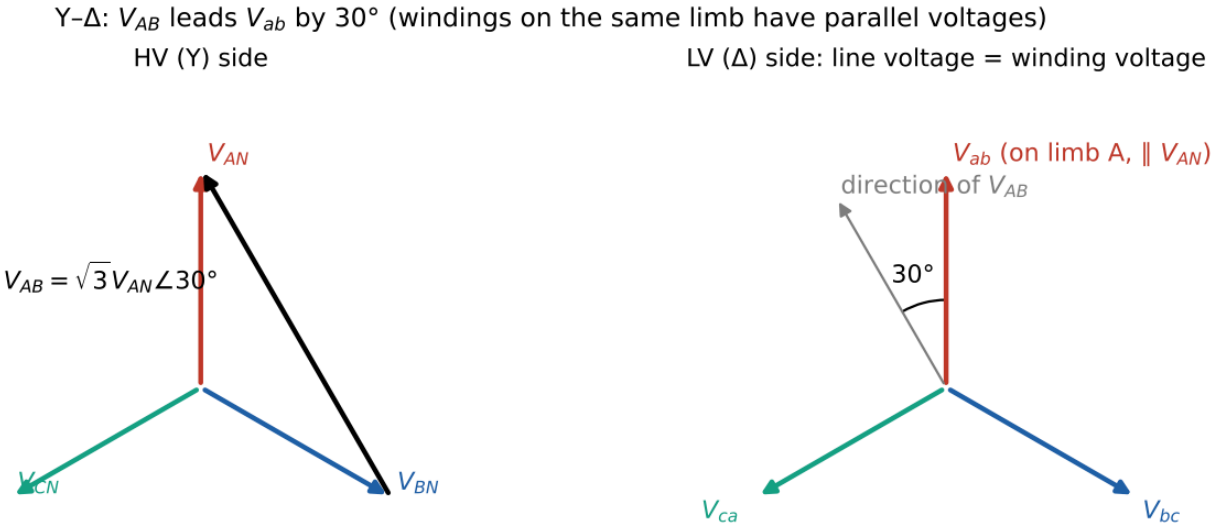


Figure 21.2. Connection diagrams; windings of the same colour are on the same core limb (or are the two windings of the same single-phase unit). The drawings show connections, not phasor directions.

3.3 Where the 30° comes from

Take Y–Δ. The LV winding on limb A is connected between lines a and b, so $\mathbf{V}_{ab}$ is parallel to the voltage of the HV winding on the same limb, $\mathbf{V}_{AN}$ (windings on one limb see the same flux). But the HV *line* voltage $\mathbf{V}_{AB} = \mathbf{V}_{AN} - \mathbf{V}_{BN} = \sqrt{3}\,\mathbf{V}_{AN}\angle 30^\circ$ (positive sequence). Therefore $\mathbf{V}_{AB}$ **leads** $\mathbf{V}_{ab}$ by 30° (Figure 21.3; Sen Fig. 2.18). The same argument gives a 30° shift for Δ–Y. Whether the shift is called +30° or −30° (LN11 slides 16–17) depends on which side is taken as reference and how terminals are labelled; nameplates use the IEC clock notation (LN11 slide 13: **Dyn11** = Δ HV, y LV with neutral n, LV lagging HV by 11 × 30° = 330°, i.e. leading by 30°). Y–Y and Δ–Δ give no shift.



The phase shift matters when transformers are paralleled (their shifts must match) and in protection; it does **not** affect magnitudes, so per-phase magnitude calculations ignore it (state that you are doing so).

3.4 Harmonics and the choice of connection (Sen §2.7)

Transformers are designed near the knee of the B–H curve, so the magnetizing current contains a third harmonic (Lesson 15). The third harmonics of the three phases are **in phase** with each other (3 × 120° = 360°), so they behave like a zero-sequence current:

- With a Δ winding they circulate inside the Δ; the flux stays sinusoidal (LN11 slide 15: Δ–Δ "no third current harmonics" in the lines).
- In Y–Y without neutral they cannot flow; the flux and phase voltages then contain third harmonics (line-to-line voltages do not). Remedy: neutral connection or a Δ **tertiary** winding.
- In a grounded-neutral Y the neutral carries $3I_{m3}$ (Example 21.12).

3.5 Open-delta (V) connection (Sen §2.6.1)

Removing one unit of a Δ–Δ bank leaves two units still supplying balanced three-phase voltages. Each remaining unit carries line current, so at rated winding current the bank can deliver $2VI\cos 30^\circ = \sqrt{3}VI$, i.e. $\sqrt{3}/3 = 57.7$ % of the full Δ–Δ rating (and 86.6 % of the two units' combined rating).

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4. Per-phase (single-phase equivalent) analysis

For a balanced system, analyze one phase of an equivalent **Y–Y** system and reconstruct the rest by 120° shifts (Sen Fig. 2.19):

1. Convert every Δ-connected element to its Y equivalent: impedances $Z_Y = Z_\Delta/3$ (LN11 slide 7: $Z_\Delta = 3Z_Y$).
2. Use phase-to-neutral voltages $V_{LL}/\sqrt{3}$ and line currents.

3. Model the bank as one single-phase transformer whose ratio is the **line-to-line voltage ratio**.
4. Refer transformer impedances to the chosen side **on a per-phase-Y basis**. Two equivalent routes (Final 2017 key): (i) take the winding impedance of one unit, convert Δ→Y if that winding is in Δ, then refer with the LL ratio squared; (ii) refer the winding impedance with the turns ratio squared to the other side's *winding*, then convert. Both give the same result.
5. Solve the single-phase circuit; convert back (×√3 for LL voltages; actual winding currents differ by √3 in Δ windings).

Regulation and efficiency follow Lesson 20 per phase; total power = 3 × per-phase power.

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5. Per-unit values (Sen §2.8 — extension used by the archive finals)

Choose base power S_b (rated VA) and base voltage V_b (rated voltage of each side); then $I_b = S_b/(\sqrt{3}V_{b,LL})$ and $Z_b = V_{b,LL}^2/S_b$ (three-phase, per-phase-Y). Per-unit value = actual/base. Key properties:

- A transformer's per-unit impedance is **the same from either side** (Sen Eq. 2.66), so no referring is needed;
- In a SC test at rated current, $Z_{pu} = V_{sc}/V_{rated}$;
- Fault current at the terminals of an infinite bus: $I_{sc,pu} = 1/|Z_{pu}|$ (plus any series impedances, also in pu on the same base).

Typical transformer values: R_{pu} 0.005–0.02, X_{pu} 0.04–0.15.

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6. Methods and common mistakes

- Always state for each quantity whether it is **line or winding (phase)**, and **which side**.
- Use the LL ratio for the per-phase equivalent; use the turns ratio only for winding quantities.
- Convert Δ impedances to Y before mixing them with line impedances.
- Feeder impedances "per phase" are line (series) impedances — already Y-compatible.
- Do not apply $\sqrt{3}$ twice: the per-phase-Y voltage is $V_{LL}/\sqrt{3}$ on both sides.

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7. Worked examples

Example 21.1 — Ratios for every connection

Given. Three single-phase 2 MVA, 10 kV/230 V units. **Find** the bank's LL voltages and LL ratio for each connection. **Solution.** Turns ratio $a = 43.48$. Y–Y: 17.32 kV/398.4 V, ratio 43.48. Δ–Δ: 10 kV/230 V, 43.48. Y–Δ: 17.32 kV/230 V, 75.31. Δ–Y: 10 kV/398.4 V, 25.10. The bank is 6 MVA in every case.

Example 21.2 — The LN11 Δ–Y bank (LN11 slides 19–22)

Given. Three single-phase 2 MVA, 10 kV/230 V, 50 Hz units in Δ–Y. Equivalent reactance of each unit referred to its LV winding $jx_1' + jx_2 = j8$ mΩ; HV source series reactance $jX_s = j0.3125$ Ω (per phase); LV feeder $jX_f = j1.5$ mΩ. **Find** (a) LL voltages and line currents on both sides; (b)

the bank rating; (c) turns and LL ratios; (d) the per-phase Y circuit referred to the LV side; (e) with the load drawing rated current at unity pf at 230 V phase (line-to-neutral) voltage, the referred source voltage and the actual source LL voltage.

Solution. (a) HV: $V_{LL} = 10$ kV (Δ), line current $6 \times 10^6/(\sqrt{3} \times 10^4) = 346.4$ A (winding 200 A). LV: $V_{LL} = \sqrt{3} \times 230 = 398.4$ V (Y), line current = winding current $= 2 \times 10^6/230 = 8696$ A. (b) 6 MVA. (c) turns ratio 43.48; LL ratio $10\,000/398.4 = 25.10$.

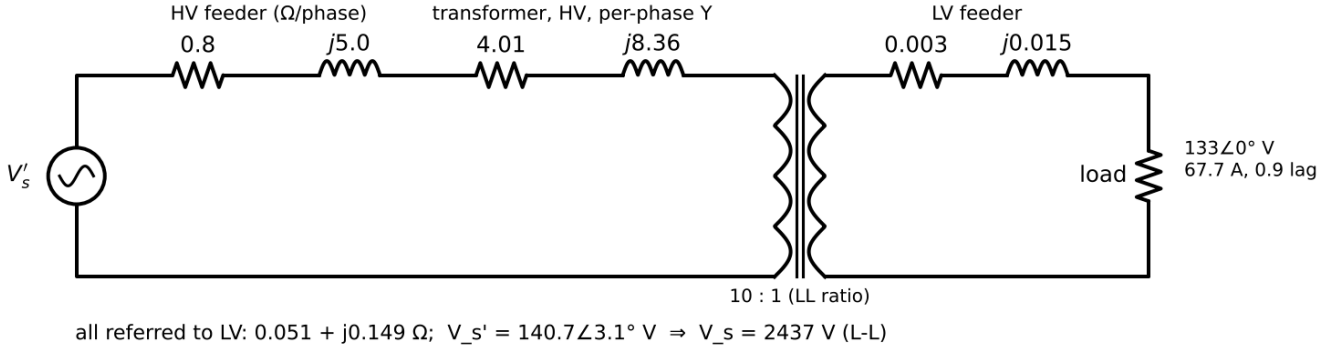
d. Per-phase Y on the LV side: the LV windings are already in Y, so the unit reactance $j8$ mΩ is directly a per-phase value. Source: $X_s' = 0.3125/25.10^2 = 0.496$ mΩ. Total $X = 0.496 + 8 + 1.5 = 10.0$ mΩ (the numbers were clearly chosen to give 10 mΩ).
e. $\mathbf{V}_s' = 230 + j(0.010)(8696) = 230 + j86.9 = 245.9\angle 20.7^\circ$ V (per phase, LV-referred). Actual source LL voltage $= 245.9 \times \sqrt{3} \times 25.10 = 10.69$ kV (equivalently $245.9 \times 10\,000/230$). **Interpretation.** At rated current the source must supply 6.9 % more than nominal voltage — almost all of it because of the transformer reactance (8 of the 10 mΩ). With $Z_{b,LV} = 398.4^2/(6 \times 10^6) = 26.5$ mΩ the transformer reactance is 0.30 pu — high for a real unit (0.05–0.15 pu is typical), but convenient for the exercise. LN11 gives no printed answers; these are ours.

Example 21.3 — Sen Example 2.7: Y–Δ bank feeding a load

Given. Three 50 kVA, 2300/230 V units form a 4000/230 V bank (HV in Y so that each winding sees $4000/\sqrt{3} \approx 2300$ V; LV in Δ). Each unit: $Z_{eq} = 0.012 + j0.016$ Ω referred to LV. Load 120 kVA, 230 V, 0.85 pf lag. **Find** winding currents, required primary LL voltage, regulation. **Solution.** LV line current $120\,000/(\sqrt{3} \times 230) = 301.2$ A; LV winding (Δ) current $301.2/\sqrt{3} = 173.9$ A; HV winding current $173.9/10 = 17.39$ A. Per unit, HV side: $Z = 100(0.012 + j0.016) = 1.2 + j1.6$ Ω; $\mathbf{V}_1 = 2300 + 17.39\angle -31.8^\circ(1.2 + j1.6) = 2332.4$ V ⇒ primary LL $= \sqrt{3} \times 2332.4 = 4040$ V; VR = 1.41 %. **Check.** Per-phase-Y on the LV side ($Z_Y = Z/3$, line current 301.2 A) gives the same regulation.

Example 21.4 — Sen Example 2.8 (LN11 slides 23–26)

Given. 3φ, 230 V (LL), 27 kVA, Y-connected, 0.9 pf lag load, fed through a feeder $0.003 + j0.015$ Ω/phase from three 10 kVA, 1330:230 V units in Y–Δ, which are fed through a feeder $0.8 + j5.0$ Ω/phase. Each unit's $Z_{eq} = 0.12 + j0.25$ Ω referred to LV. **Find** the source LL voltage for 230 V at the load. **Solution.** Unit impedance referred to its HV winding: $(1330/230)^2(0.12 + j0.25) = 4.01 + j8.36$ Ω; the HV windings are in Y, so this is per-phase Y. Equivalent Y–Y ratio $a' = \sqrt{3} \times 1330/230 = 10.0$ (= LL ratio). Refer all to LV: $R = (0.80 + 4.01)/100 + 0.003 = 0.051$ Ω, $X = (5 + 8.36)/100 + 0.015 = 0.149$ Ω (Figure 21.4). Load phase voltage $133\angle 0^\circ$ V, current $27\,000/(3 \times 133) = 67.7$ A at $-25.8°$. $\mathbf{V}_s' = 133 + 67.7\angle -25.8^\circ(0.051 + j0.149) = 140.5\angle 3.1^\circ$ V ⇒ $V_s = 1405$ V per phase on the HV side, **2437 V line-to-line** (Sen: 2437 V).



Example 21.5 — Short-circuit test, per-unit and fault current (Final Fall 2014 Q1)

Given. (exams/archive/Final/Final - Fall2014.pdf p. 1.) 500 kVA, 24 000/2400 V, three-phase Y/Δ transformer fed from a 24 kV infinite bus through a line of $j0.8\ \Omega$/phase; load on the LV ("transformer") bus. (a) A three-phase SC test at rated current from the HV side (LV shorted): $P_{in} = 6$ kW, 1936 V LL. Find the per-phase impedance on HV and LV sides. (b) Per-unit values. (c) A three-phase fault at the transformer (LV) bus: fault current and MVA, using per-unit. (d) Effect of a practical (non-infinite) source with $X_s = 0.1\ \Omega$.

Solution. (a) Rated HV line current $500\,000/(\sqrt{3} \times 24\,000) = 12.03$ A. Per-phase Y on HV: $|Z| = (1936/\sqrt{3})/12.03 = 92.93\ \Omega$; $R = 6000/(3 \times 12.03^2) = 13.82\ \Omega$; $X = 91.89\ \Omega$. LV per-phase Y: ÷(LL ratio 10)² ⇒ $0.1382 + j0.9189\ \Omega$ (the actual Δ winding impedance is three times that).

b. $Z_{b,HV} = 24\,000^2/500\,000 = 1152\ \Omega \Rightarrow Z_{pu} = 0.0120 + j0.0798$, $|Z_{pu}| = 0.0807$ (= 1936/24 000 ✓). The same per-unit value results from the LV side with $Z_{b,LV} = 11.52\ \Omega$.

c. Line: $0.8/1152 = j0.000694$ pu. $I_{sc} = 1/|j0.000694 + 0.0120 + j0.0798| = 12.29$ pu ⇒ $12.29 \times 120.3 = 1479$ A on the LV side (147.9 A on the HV lines); fault MVA = $12.29 \times 0.5 = 6.15$ MVA.

d. An extra $X_s = 0.1\ \Omega$ (0.0000868 pu) lowers the fault current only to 12.28 pu: the transformer impedance dominates. In general a weaker source (larger X_s) reduces the fault current and MVA.

Example 21.6 — Δ/Y bank between a utility grid and a distribution bus (Final Fall 2017 Q1)

Given. (exams/archive/Final/Final - Fall2017.pdf pp. 2–3, key p. 13.) Three 5 MVA, 66 kV/3.8 kV units in Δ/Y deliver rated power at rated voltage and unity pf to a 6.6 kV bus. Grid $X_s = 0.20\ \Omega$ per-phase-Y (HV); each unit $X'_{tr} = 2.4\ \Omega$ on the LV side; feeder $X_f = 0.40\ \Omega$ (LV). **Find** (A) turns ratio, LV LL rating, LL ratio, power rating; (B) the per-phase-Y circuit referred to HV and the phasor diagram.

Solution. (a) $n = 66/3.8 = 17.37$. (b) $V_{2,LL} = \sqrt{3} \times 3.8 = 6.58 \approx 6.6$ kV. (c) LL ratio $= 66/6.6 = 10$ (≠ n). (d) 15 MVA. (e) Referred to HV with the LL ratio: $X_{tr} = 2.4 \times 10^2 = 240\ \Omega$, $X_f = 0.4 \times 10^2 = 40\ \Omega$; total $0.2 + 240 + 40 = 280.2\ \Omega$ (key ✓). Load: $V'_L = 6.6 \times 10/\sqrt{3} = 38.1$ kV per phase, $I'_L = 15 \times 10^6/(3 \times 38.1 \times 10^3) = 131.2$ A in phase with it. (f) Phasors: $\mathbf{V}'_2 = \mathbf{V}'_L + jX'_f\mathbf{I}'$, $\mathbf{V}_1 = \mathbf{V}'_2 + jX_{tr}\mathbf{I}'$, $\mathbf{V}_s = \mathbf{V}_1 + jX_s\mathbf{I}'$, each drop at +90° to the current (key sketch). Beyond the key: $|V_s| = 52.95$ kV per phase (91.7 kV LL). Observation: the transformer reactance is $240/(66^2/15) = 0.83$ pu, unrealistically large (real units: ~0.1 pu), so the required source voltage is far above 66 kV. The method is what the question tests.

Example 21.7 — Δ–Y bank built from tested units, feeding homes (Jan 2022 Final Q1(c)–(h))

Given. (exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf pp. 2–5.) Three units of Lesson 19 Ex. 19.3 (10 kVA, 2500/250 V; per HV winding $Z_{eq} = 5 + j19.36\ \Omega$) in Δ–Y; HV feeder $5 + j21\ \Omega$/phase; balanced load 24 kW (total), 0.8 pf **leading**, at 430 V LL; shunt branch ignored. **Find** (c) the connection and its advantage; (d) LL ratio; (e) per-phase circuit; (f) V_{an} and I_a; (g) transformer VR; (h) P, Q and pf at the sending end of the feeder.

Solution. (c) HV windings in Δ, LV in Y with neutral: the LV neutral serves single-phase household loads, the Δ provides a path for triplen harmonic currents, and the HV side needs no neutral. (d) $2500/(\sqrt{3} \times 250) = 2500/433 = 5.774$.

e. Per-phase Y referred to HV: transformer $Z_\Delta/3 = 1.667 + j6.455\ \Omega$ in series with the feeder $5 + j21\ \Omega$; ideal ratio 5.774.
f. $\mathbf{V}_{an} = 430/\sqrt{3} = 248.3\angle 0^\circ$ V; $|I_a| = 24\,000/(3 \times 248.3 \times 0.8) = 40.28$ A, $\mathbf{I}_a = 40.28\angle +36.87^\circ$ A.
g. HV-referred: $V_2' = 248.3 \times 5.774 = 1433.3$ V; $I' = 40.28/5.774 = 6.977$ A; $\mathbf{V}_1' = 1433.3 + 6.977\angle 36.87^\circ(1.667 + j6.455) = 1416.3\angle 1.74^\circ$ V ⇒ VR = **−1.19 %** (leading load).
h. $\mathbf{V}_s = \mathbf{V}_1' + \mathbf{I}'(5 + j21)$ ⇒ $|V_s| = 2369$ V LL. $S_s = 3\mathbf{V}_s\mathbf{I}'^* = 24.97$ kW $-j13.99$ kvar ⇒ pf 0.872 leading. **Check.** $P_s = 24\,000 + 3I'^2(1.667 + 5)$ W ✓. *(If the fallback $3 + j15\ \Omega$ per phase given in the paper were used as a per-phase-Y value, VR would be −2.98 %.)*

Example 21.8 — Δ–Y distribution transformer: efficiency, SC test, feeder (Jan 2023 Final Q1)

Given. (exams/archive/20221/EE361_Final_2022_Q4_Solution.pdf pp. 2–5.) 12 kV/400 V, Δ–Y, 520 kVA single-core unit; winding impedances $Z_{LV} = (1.2 + j12) \times 10^{-3}\ \Omega$, $Z_{HV} = 3.96 + j39.6\ \Omega$. **Find** (a) the phase (winding) equivalent impedance referred to HV; (b) maximum efficiency and the current at which it occurs if $P_{core} = 3$ kW; (c) SC test from HV at nominal current — why HV, why neglect the shunt, source LL voltage and line current. Part II: rated resistive load at rated voltage via a feeder $2.6 + j6\ \Omega$/phase: (a) VR; (b) source P, Q, pf; (c) losses and overall efficiency.

Solution. (a) Turns ratio per winding $12\,000/(400/\sqrt{3}) = 51.96$, squared 2700: $Z_{eq} = 3.96 + j39.6 + 2700(1.2 \times 10^{-3} + j0.012) = 7.2 + j72\ \Omega$ per Δ winding.

b. Rated winding current $520\,000/(3 \times 12\,000) = 14.44$ A; $P_{cu,FL} = 3 \times 14.44^2 \times 7.2 = 4507$ W. $x = \sqrt{3000/4507} = 0.816$ ⇒ maximum efficiency at **81.6 %** of rated current, $\eta_{max} = 0.816 \times 520/(0.816 \times 520 + 6) = 98.6$ % (unity pf).
c. HV side: smaller current, a measurable fraction of HV voltage, safer high-current shorting on LV. The shunt branch is negligible because the test voltage is only a few percent of rated (Lesson 19). Source: winding current 14.44 A ⇒ line current $\sqrt{3} \times 14.44 = 25.0$ A; LL voltage = Δ winding voltage $= 14.44 \times |7.2 + j72| = 1045$ V (8.7 % of 12 kV). Part II (per-phase Y on HV: transformer $2.4 + j24\ \Omega$): (a) $V_2' = 6928$ V, $I = 25.02$ A at unity pf: $|V_1'| = 7014$ V ⇒ VR = 1.24 %. (b) $P = 520 + 3(25.02^2)(2.4 + 2.6) \times 10^{-3} = 529.4$ kW; $Q = 3(25.02^2)(24 + 6) = 56.3$ kvar; pf = 0.994 lagging. (c) Losses 9.39 kW (core loss ignored as instructed), η = 98.2 %; source LL voltage 12.29 kV.

Example 21.9 — The handwritten LN11 single-line example (LN11 slide 27)

Given. Utility grid (reactance X_s, value not given), cable $X_c = 1\ \Omega$, 3 MVA Δ/Y 10 kV/0.4 kV transformer with $X_{eq} = 10/3\ \Omega$ per-phase-Y (taken as HV-side value), LV feeder $X_f = 1$ mΩ, three-phase load. **Find** the per-phase-Y circuit on the HV side. **Solution.** LL ratio 25; refer the feeder: $1 \times 10^{-3} \times 25^2 = 0.625\ \Omega$. Per-phase-Y circuit on HV: $E_s/\sqrt{3}$ — jX_s — $j1\ \Omega$ — $j3.333\ \Omega$ — $j0.625\ \Omega$ — load (referred, voltage ×25, current ÷25). With $Z_{b,HV} = 10^2/3 = 33.3\ \Omega$, the transformer is 0.10 pu — a realistic value. (Numerical results need the missing X_s and load data; the slide asks only for the circuit.)

Example 21.10 — Open-delta capacity

Two 100 kVA units in open-delta can supply $\sqrt{3} \times 100 = 173$ kVA without overload — 57.7 % of the 300 kVA Δ–Δ bank, 86.6 % of their own 200 kVA.

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8. Optional practice

P21.1. Three 1φ 100 kVA, 7200/240 V units are connected Y–Δ. Find the bank's rating, LL voltages, LL ratio and the line currents at rated load.

▼ **Solution**
300 kVA; HV $\sqrt{3} \times 7200 = 12.47$ kV LL; LV 240 V LL; LL ratio 51.96. HV line current $300\,000/(\sqrt{3} \times 12\,470) = 13.9$ A (= winding current); LV line current $300\,000/(\sqrt{3} \times 240) = 722$ A (winding $100\,000/240 = 417$ A).

P21.2. In Example 21.4, what source LL voltage is needed if the load power factor is 0.9 leading instead?

▼ **Solution**
$\mathbf{V}'_s = 133 + 67.7\angle 25.8^\circ(0.051 + j0.149) \approx 131.5 + j10.5$ V ⇒ $|V'_s| = 131.9$ V ⇒ LL $= 131.9 \times 10.02 \times \sqrt{3} = 2296$ V (below the 2304 V nominal $1330\sqrt{3}$). A leading load needs less source voltage.

P21.3. A 1 MVA, 11 kV/415 V transformer has $Z_{pu} = 0.05$. Find the fault current on the LV side for a bolted three-phase fault at its terminals (infinite bus).

▼ **Solution**
$I_b = 10^6/(\sqrt{3} \times 415) = 1391$ A; $I_{sc} = 1391/0.05 = 27.8$ kA (20 pu).

P21.4 (concept). Why is a Y–Y transformer rarely used without a neutral connection or tertiary winding?

▼ **Solution**
The third-harmonic magnetizing currents of the three phases are in phase; without a neutral or a Δ winding they cannot flow, so the flux becomes non-sinusoidal and the phase voltages contain large third-harmonic components (neutral-point oscillation), stressing insulation and distorting phase-to-neutral voltages. A Δ tertiary provides a path for those currents (Sen §2.7).

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9. Summary

Relation	Validity
$S = 3V_{ph}I_{ph} = \sqrt{3}V_{LL}I_L$	any connection, balanced
LL ratio: Y–Y, Δ–Δ = a; Y–Δ = $\sqrt{3}a$; Δ–Y = $a/\sqrt{3}$	a = winding turns ratio
Y–Δ, Δ–Y: 30° shift between LL voltages	positive sequence
$Z_Y = Z_\Delta/3$; per-phase ratio = LL ratio	balanced per-phase analysis
Z_{pu} same on both sides; $I_{sc} = 1/Z_{pu}$ (infinite bus)	per-unit (extension)
Open-delta = 57.7 % of Δ–Δ	two of three units

10. Vocabulary

English	Türkçe
three-phase transformer; transformer bank	üç fazlı transformatör; transformatör grubu (bankı)

English	Türkçe
wye (star) / delta connection	yıldız / üçgen bağlantı
line-to-line voltage; phase voltage	hat (fazlar arası) gerilimi; faz gerilimi
phase shift; vector group	faz kayması; bağlantı grubu
per-phase equivalent circuit	faz başına eşdeğer devre
tertiary winding	üçüncül sargı
open-delta (V) connection	açık üçgen (V) bağlantı
per-unit system	birim değer (pu) sistemi
fault current; short-circuit MVA	arıza akımı; kısa devre gücü

11. Sources

- LN11 viewer pp. 3–50 (slides 2–29): why three-phase, Y/Δ relations, power, impedance conversion, three-limb core, connections and phase shifts, nameplate Dyn11, turns vs LL ratio, 2 MVA bank example, Sen Ex. 2.8, handwritten example. Notes.
- Sen §2.6–2.8 pp. 66–84 (viewer 90–108): connections Fig. 2.17, phase shift Fig. 2.18, single-phase equivalent Fig. 2.19, Eqs. 2.39–2.45, Examples 2.7–2.8, V connection, three-phase core Fig. 2.21, harmonics §2.7, per-unit §2.8.
- Final Fall 2014 Q1 p. 1; Final Fall 2017 Q1 pp. 2–3 and key p. 13; Jan 2022 Final Q1 pp. 2–5; Jan 2023 Final Q1 pp. 2–5.
- Verification: `lesson21_three_phase_transformers.py`.
- More solved archive problems: Ch II–III archive collection §5 (Final Fall 2016 Q1, per-unit transformer).

Chapters II–III — remaining archive problems with full solutions

Assistant-written worked solutions, 2 November 2026. These archive questions were inventoried during the source audit but are not solved inside Lessons 10–21. Original papers are in `sources/exams/archive/` (page = PDF viewer page). Figures from the papers are restated, not copied; where a key exists it was read visually and every number recomputed. Numbers: `assistant/calculations/full-semester/collection_ch2_ch3_archive.py` (34 checks). Archive papers show past problem types, not the 2026 exam scope.

Contents:

1. MT I Fall 2015 Q2 — trapezoidal flux on a hysteretic core (Lessons 15–16)
2. MT I Fall 2015 Q3(a)–(c) — voltage, frequency and saturation (Lesson 15)
3. MT I Fall 2019 Q2(e) — why inductors have air gaps (Lessons 13–14)
4. MT I Fall 2019 Q3 — 1.25 MVA distribution transformer (Lessons 18–20)
5. Final Fall 2016 Q1 — per-unit transformer, waveforms and efficiency (Lessons 20–21)
6. Final Fall 2017 Q2 Part I — EMF equation and power rating (Lessons 15, 17, 20)
7. MT1 recitation (Dr. Ermiş) — V/f excitation, two regulation examples, HW1 three-leg core (Lessons 12, 15, 18, 20)

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1. MT I Fall 2015 Q2 — trapezoidal flux on a hysteretic core

Source. `exams/archive/MT/MT I - Fall2015.pdf` p. 3 (question, Fig. 1a–c) and p. 5 (answer sheet template). No key supplied. Related lessons: 15 (Faraday, waveforms, harmonics), 16 (hysteresis loss = loop area).

Given (restated from the figures). Core area A = 10 cm², N = 100, mean length ℓ = 40 cm; leakage and winding resistance neglected.

- B–H loop (Fig. 1b), a parallelogram with corners (H, B) = (−7.5 kA/m, −1.2 T), (0, −0.6 T), (7.5 kA/m, 1.2 T), (0, 0.6 T). The ascending branch is (−7.5, −1.2) → (0, −0.6) → (7.5, 1.2); the descending branch is (7.5, 1.2) → (0, 0.6) → (−7.5, −1.2). Both cross B = 0 at $H = \pm 2.5$ kA/m.
- Flux (Fig. 1c), a trapezoid over T = 20 ms: rises linearly from 0 (point a, t = 0) through 0.6 mWb (b, 1 ms) to 1.2 mWb (c, 2 ms); flat to d (8 ms); falls through e (0.6 mWb, 9 ms), f (0, 10 ms), g (−0.6 mWb, 11 ms) to h (−1.2 mWb, 12 ms); flat to j (18 ms); rises through k (−0.6 mWb, 19 ms) to q (0, 20 ms).

Find. (a) frequency; (b) Φ–B and H–i relations; (c) current at the marked points; (d) $e(t)$ and its first harmonic; (e) show $E_{1,rms} = 22.2 \times 10^3\, \phi_{1mx} = 22.2\, B_{1mx}$; (f) core loss.

Solution.

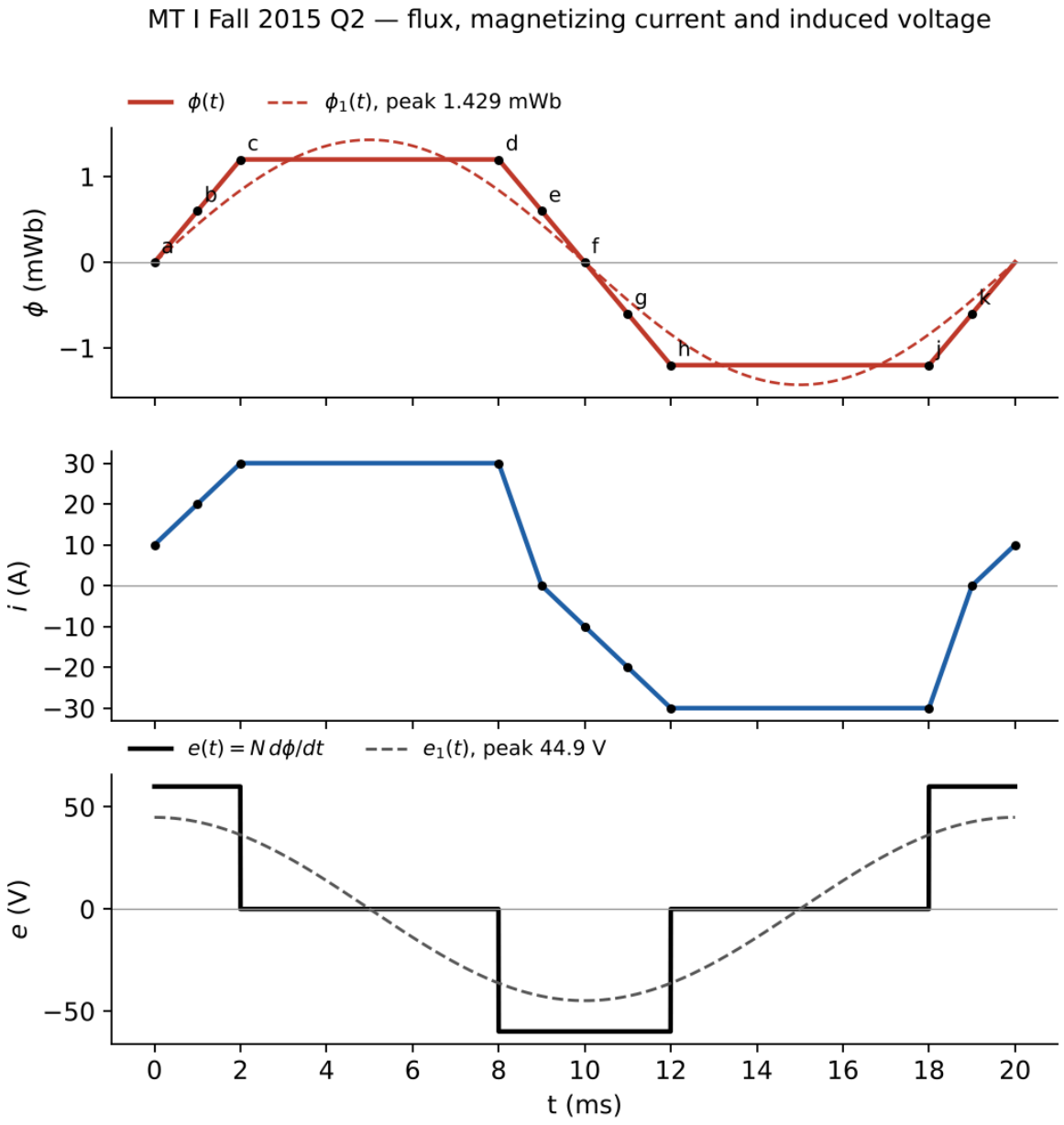
a. One period is 20 ms ⇒ $f = 1/T$ = **50 Hz**.

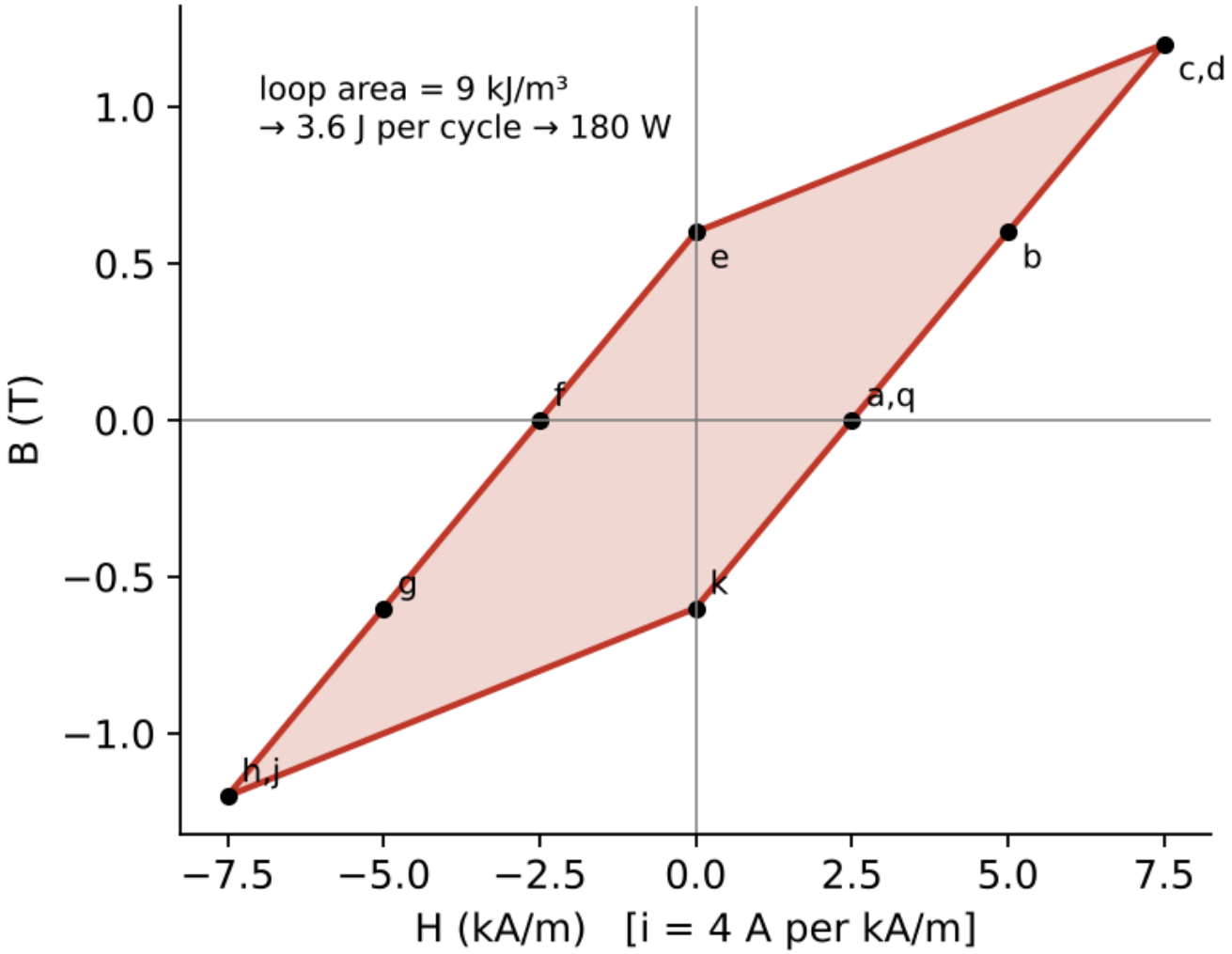
b. Uniform flux in the core: $\Phi = BA = 10^{-3} B$ (Φ in Wb, B in T), so 1.2 T ↔ 1.2 mWb — the two vertical scales coincide. Ampère's law around the mean path: $H\ell = Ni \Rightarrow i = H\ell/N = 0.004\, H$, i.e. **4 A per kA/m**: 2.5 kA/m ↔ 10 A, 5 kA/m ↔ 20 A, 7.5 kA/m ↔ 30 A.

c. Without hysteresis the current would follow the flux on one curve. With hysteresis we must know **which branch** is active: rising flux → ascending branch, falling flux → descending branch, and during the flat parts the operating point stays where it is.

Point	t **(ms)**	**Φ (mWb)**	**Branch**	H **(kA/m)**	i **(A)**
a	0	0	ascending	2.5	**10**
b	1	0.6	ascending	5.0	**20**
c, d	2–8	1.2	tip	7.5	**30**
e	9	0.6	descending	0	**0**
f	10	0	descending	−2.5	**−10**
g	11	−0.6	descending	−5.0	**−20**
h, j	12–18	−1.2	tip	−7.5	**−30**
k	19	−0.6	ascending	0	**0**
q	20	0	ascending	2.5	**10**

Between the points the current varies linearly (each branch segment is straight and Φ is linear in time). Note that $i \neq 0$ when Φ = 0 (points a, f, q): this is the **coercive** MMF; and Φ ≠ 0 when i = 0 (points e, k): **remanence**. The current is not symmetric in time with the flux — it leads it, which is the waveform signature of hysteresis loss.


MT I Fall 2015 Q2 — flux, magnetizing current and induced voltage
$\phi(t)$
$\phi_1(t)$, peak 1.429 mWb
ϕ (mWb)
a
b
c
d
e
f
g
h
j
k
i (A)
$e(t) = N\,d\phi/dt$
$e_1(t)$, peak 44.9 V
e (V)
t (ms)



d. $e = N\,d\Phi/dt$ (passive convention, winding resistance zero). On the ramps $|d\Phi/dt| = 1.2\ \text{mWb}/2\ \text{ms} = 0.6\ \text{Wb/s} \Rightarrow |e| = 60\ \text{V}$; on the flats $e = 0$:

- e = +60 V for 0–2 ms and 18–20 ms (one 4 ms pulse centred on t = 0),
- e = 0 for 2–8 ms and 12–18 ms,
- e = −60 V for 8–12 ms.

This is a quasi-square wave of amplitude E = 60 V and half-width $\beta = \omega \times 2\ \text{ms} = 36^\circ$ about t = 0 and $t = T/2$. It is even in t, so its fundamental is a cosine:

$$a_1 = \frac{2}{T}\int_0^T e(t)\cos\omega t\,dt = \frac{4E}{\pi}\sin\beta = \frac{4(60)}{\pi}\sin 36^\circ = 44.90\ \text{V}.$$

So $e_1(t) = 44.9\cos(100\pi t)$ V; $E_{1,rms}$ = 31.75 V. (Numerically confirmed by FFT.)

e. The fundamental flux is the integral of e_1: $\phi_1(t) = \dfrac{a_1}{N\omega}\sin\omega t$, so $\phi_{1mx} = a_1/(N\omega)$ and

$$E_{1,rms} = \frac{a_1}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} f N \phi_{1mx} = 4.44\,f N\phi_{1mx} = 4.44(50)(100)\phi_{1mx} = 22.2\times 10^3\,\phi_{1mx}.$$

With $\phi_{1mx} = B_{1mx}A = 10^{-3}B_{1mx}$: $E_{1,rms} = 22.2\,B_{1mx}$. ∎ Numerically $\phi_{1mx} = 44.90/(100\times 100\pi)$ = 1.429 mWb (B_1 = 1.429 T). Check from the flux trapezoid directly: its Fourier fundamental is $\frac{4\Phi_m}{\pi}\frac{\sin\beta}{\beta} = \frac{4(1.2)}{\pi}\frac{0.5878}{0.6283}$ = 1.429 mWb ✓. The fundamental is *larger* than the 1.2 mWb trapezoid peak — the dashed sine in the figure.

f. Hysteresis loss per cycle = loop area × core volume. Shoelace on the four corners: area = 9 kA/m·T = **9 kJ/m³**. Volume $A\ell = 10^{-3} \times 0.4 = 4 \times 10^{-4}$ m³ ⇒ 3.6 J per cycle ⇒ $P_c = 3.6 \times 50$ = **180 W**. *Independent check:* average of $e\,i$ over the period. On 0–1 ms e = 60 V and i rises 10→20 A (energy 60 × 15 × 1 ms = 0.9 J); 1–2 ms: 1.5 J; 8–9 ms: −0.9 J; 9–10 ms: +0.3 J; 10–11 ms: 0.9 J; 11–12 ms: 1.5 J; 18–19 ms: −0.9 J; 19–20 ms: +0.3 J. Total 3.6 J ✓. The negative intervals are energy returned from the field to the source.

Interpretation. The flux shape is fixed by the voltage; the current shape is fixed by the B–H loop. A flat-topped flux gives a pulsed voltage whose fundamental flux exceeds the trapezoid peak by 19 %.

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2. MT I Fall 2015 Q3(a)–(c) — voltage, frequency and saturation

Source. `exams/archive/MT/MT I - Fall2015.pdf` p. 4 (Figs. 1–2). Parts (d)–(e) are solved in Lesson 19, Example 19.4. No key supplied.

Given. 220 V, 50 Hz sinusoidal source on N = 412 turns; secondary open; core area 20 cm². B–H (Fig. 2): linear from the origin to the knee (3000 A/m, 1.7 T), then nearly flat (saturation). Core length not given.

(a) Assumptions: winding resistance and leakage neglected, so the applied voltage equals the induced EMF; flux uniform over the area. Then $V = 4.44 f N B_{max} A$:

$$B_{max} = \frac{220}{4.44(50)(412)(20 \times 10^{-4})} = \mathbf{1.20\ T}.$$

(b) The voltage is sinusoidal, so the flux (its integral) is sinusoidal: $B(t) = 1.20 \sin(\omega t)$ T, lagging the voltage by 90°. Since 1.20 T < 1.7 T, the core stays on the linear part of the B–H curve and the magnetizing current is **sinusoidal and in phase with the flux** (the given curve has no hysteresis loop; with hysteresis it would be slightly distorted and lead the flux). Its amplitude cannot be computed because the mean length is not given.

(c) At 30 Hz with the same 220 V: $B_{max} = 1.20 \times 50/30$ = **2.00 T**. The voltage still *forces* this flux (Faraday's law), but 2.0 T is beyond the 1.7 T knee: the core saturates for part of each half-cycle. In saturation a small increase in B needs a very large increase in H, so the magnetizing current becomes **peaky** (large narrow pulses around the flux peaks, rich in odd harmonics — 3rd, 5th, …) and its rms value rises sharply. Physically: V/f sets flux, so reducing f at constant V over-fluxes the core; drives keep V/f constant (Section 7.1 below). See Lesson 15, §3.3–4.

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3. MT I Fall 2019 Q2(e) — design an ungapped inductor with the same L and saturation current

Source. `exams/archive/MT/MT I - Fall2019.pdf` p. 4 (Page 8 of 12). Parts (a)–(d) are Lesson 15, Example 15.7; (f)–(g) are Lesson 18 Ex 18.4 and Lesson 13 P13.4. "There is no single answer"; we state assumptions.

Original design (from Example 15.7). A = 400 mm², 218 mm of core + 2 mm gap, N = 200, knee (250 A/m, 1.6 T): i_{sat} = 13.0 A, L = 9.84 mH, stored energy at the knee $W = \frac{1}{2} L i_{sat}^2$ = **0.832 J**. Where is it stored? Core: $\frac{1}{2} B H V_c = \frac{1}{2}(1.6)(250)(8.72 \times 10^{-5})$ = 0.017 J. Gap: $\frac{1}{2} B (B/\mu_0) V_g$ = 0.815 J. **98 % of the energy is in the 2 mm gap.**

The key constraint. Without a gap, all of W must sit in the steel. At most $\frac{1}{2}B_{sat}H_{sat} = \frac{1}{2}(1.6)(250) = 200$ J/m³ can be stored before saturation, so the core volume must be at least

$$V_c = \frac{0.832}{200} = 4.16 \times 10^{-3}\ \text{m}^3 \approx 48 \times \text{ the original core.}$$

Equivalently: $N\Phi_{sat} = Li_{sat} \Rightarrow NA = Li_{sat}/B_{sat}$ = 0.0800 m², and $Ni_{sat} = H_{sat}\ell \Rightarrow \ell = Ni_{sat}/250$ = 0.0520 N. Any N works, with $V_c = A\ell$ fixed:

N	A	ℓ	**steel mass (7650 kg/m³)**
200	400 mm²	10.4 m	31.8 kg
50	1600 mm² (4 × 4 cm)	2.60 m	31.8 kg
20	4000 mm² (6.3 × 6.3 cm)	1.04 m	31.8 kg

Check (N = 20): $L = N^2\mu A/\ell = 400 \times (1.6/250) \times 0.004/1.040$ = 9.84 mH ✓; $i_{sat} = 250 \times 1.040/20$ = 13.0 A ✓.

Comparison. The original gapped inductor uses 0.67 kg of steel; any ungapped equivalent needs ~32 kg — physically possible but absurd, with large core loss and cost. The gap is preferable because (i) air stores energy at $B^2/2\mu_0$ — thousands of times more per volume than steel at the same B; (ii) the gap reluctance dominates, so L is set by geometry, not by the uncertain, temperature-dependent μ of the steel; (iii) the knee current is controlled by the gap length. This is exactly why inductors have gaps and transformers do not (Lesson 13 §2.3).

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4. MT I Fall 2019 Q3 — 1.25 MVA, 34.5 kV / 230 V distribution transformer

Source. `exams/archive/MT/MT I - Fall2019.pdf` pp. 6–8. No key supplied. Lessons 18, 20.

Given. Single-phase 1.25 MVA, 34.5 kV / 230 V. R_1 = 9 Ω, R_2 = 0.0004 Ω, X_1 = 36 Ω, X_2 = 0.0016 Ω, R_c = 238 kΩ, X_m = 15 kΩ (HV side); shunt branch moved to the primary terminals.

(a) $a = 34\,500/230 = 150$, a^2 = 22 500. Referred to HV: $R_2' = 0.0004 \times 22\,500$ = 9 Ω, X_2' = 36 Ω. So R_{eq} = **18 Ω**, X_{eq} = **72 Ω**, with R_c = 238 kΩ ∥ jX_m = j15 kΩ across the 34.5 kV terminals (approximate circuit of Lesson 18 §4). Note the design symmetry $R_1 = R_2'$, $X_1 = X_2'$.

(b) No load: only the shunt branch draws power. $P_0 = V^2/R_c = 34\,500^2/238\,000$ = **5.00 kW**; $Q_0 = V^2/X_m$ = 79.35 kvar; $S_0 = \sqrt{P_0^2 + Q_0^2}$ = **79.5 kVA** (6.4 % of rating — mostly reactive magnetizing power).

(c) Resistive load, I_2 = 5250 A ⇒ I_2' = 5250/150 = 35 A in phase with $\mathbf{V}_2'$ (reference). KVL: $\mathbf{V}_1 = V_2' + 35(18 + j72) = (V_2' + 630) + j2520$. With $|V_1|$ = 34 500 V:

$$V_2' + 630 = \sqrt{34\,500^2 - 2520^2} = 34\,407.8 \ \Rightarrow\ V_2' = 33\,777.8\ \text{V}, \qquad V_2 = V_2'/150 = \mathbf{225.2\ V}.$$

Phasor diagram: $\mathbf{V}_2'$ and $\mathbf{I}_2'$ along the real axis, $R_{eq}\mathbf{I}_2'$ (630 V) added in phase, $jX_{eq}\mathbf{I}_2'$ (2520 V) at +90°, closing on $\mathbf{V}_1$ at angle δ = 4.2° ahead.

(d) $P_{out} = 225.19 \times 5250$ = 1182.2 kW; copper loss $35^2 \times 18$ = 22.05 kW; core loss 5.00 kW (rated primary voltage). $\eta = 1182.2/(1182.2 + 22.05 + 5.00)$ = **97.8 %**.

(e) Seen from the primary, the load is resistive but the transformer adds series reactance (X_{eq}) and a shunt magnetizing reactance — both absorb reactive power. So the primary current lags the primary voltage: **lagging** power factor (exact calculation: pf = 0.991 lagging).

(f) Short circuit at the secondary with 34.5 kV applied: $I'_{sc} = 34\,500/|18 + j72|$ = 34 500/74.22 = 464.9 A on the HV side, i.e. $I_{sc} = 150 \times 464.9$ = **69.7 kA** on the LV side (12.8 × rated 5435 A). Copper loss $464.9^2 \times 18$ = **3.89 MW**, 176 × the 22 kW of part (d), dissipated in the windings. Consequences: the winding temperature rises within seconds to damaging levels, and the electromagnetic forces ($\propto I^2$) on the windings are ~160 × normal — mechanical deformation. Protection (fuses/breakers) must clear the fault within a short time. The series impedance (here 0.078 pu) is what limits the fault current: $I_{sc} \approx I_{rated}/Z_{pu}$.

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5. Final Fall 2016 Q1 — 1 MVA, 33 kV / 400 V transformer in per unit

Source. `exams/archive/Final/Final - Fall2016.pdf` pp. 2–3 with handwritten key. **Key confirmed.** Lessons 20, 21 §5 (per unit).

Given. Single-phase 1 MVA, 33 kV / 400 V, 50 Hz; X_{eq} = 0.09 pu, R_{eq} = 0.02 pu; shunt branch neglected.

(a) LV bases: S_b = 1 MVA, V_b = 400 V, $I_b = S_b/V_b$ = 2500 A, $Z_b = V_b/I_b$ = 0.16 Ω. Actual LV values: $R_{eq} = 0.02 \times 0.16$ = **3.2 mΩ**, X_{eq} = **14.4 mΩ**. Circuit: $\mathbf{V}'_1$ — R_{eq} — jX_{eq} — $\mathbf{V}_2$, current $\mathbf{I}_2$.

(b) Load 0.8 pu at unity pf with V_2 = 400 V: $I_2 = 0.8 \times 2500$ = 2000 A in phase with $\mathbf{V}_2$.

$$\mathbf{V}'_1 = 400 + 2000(0.0032 + j0.0144) = 406.4 + j28.8 = \mathbf{407.4\angle 4.05^\circ}\ \mathbf{V}.$$

In per unit: $1 + 0.8(0.02 + j0.09) = 1.016 + j0.072 = 1.0185\angle 4.05^\circ$ ✓. Phasor diagram: $\mathbf{V}_2$ and $\mathbf{I}_2$ together on the reference axis, small $R_{eq}\mathbf{I}_2$ in phase, $jX_{eq}\mathbf{I}_2$ vertical; load angle δ = 4.05°.

(c) Waveforms: $v_2(t) = 400\sqrt{2}\sin\omega t$ (peak 565.7 V), $i_2(t) = 2000\sqrt{2}\sin\omega t$ (peak 2828 A, in phase), $v'_1(t) = 407.4\sqrt{2}\sin(\omega t + 4.05^\circ)$ (peak 576.2 V). v'_1 leads v_2 by $4.05/360 \times 20$ ms = **0.225 ms**.

(d) Only copper loss: $P_{cu} = 2000^2 \times 0.0032$ = 12.8 kW; P_{out} = 800 kW ⇒ $\eta = 800/812.8$ = **98.4 %**. Full load, unity pf: $P_{cu} = 2500^2 \times 0.0032$ = 20 kW ⇒ $\eta = 1000/1020$ = **98.0 %** (per unit: $1/(1 + 0.02)$). Comment: with no core loss modelled, efficiency falls monotonically with load ($P_{cu} \propto I^2$ grows faster than $P_{out} \propto I$). In a real transformer the core loss makes efficiency peak where copper loss equals core loss (Lesson 20 §3.3).

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6. Final Fall 2017 Q2 Part I — EMF equation and power rating

Source. `exams/archive/Final/Final - Fall2017.pdf` p. 5 (question), p. 15 (key). Part II (DC machine coil) is in the Chapter IV collection.

(a) With leakage neglected the same flux $\phi = \Phi_m \cos\omega t = B_{max}A\cos\omega t$ links both windings. $e_2 = -N_2\, d\phi/dt = N_2\omega B_{max}A\sin\omega t$, so $E_{2,max} = 2\pi f N_2 B_{max} A$ and

$$E_{2,rms} = \frac{2\pi}{\sqrt{2}} f N_2 B_{max} A = 4.44\, f N_2 B_{max} A.$$

(The key's derivation is identical.)

(b) Rating $S = E_2 I_2$, where E_2 is limited by B_{max} (saturation, core loss) and I_2 by heating (current density in the copper). One change at a time:

- **i. Doubling f.** At the same B_{max} the EMF doubles for the same turns and the same current ⇒ rating **doubles** (key: "power is increased"). Caveat: core loss per kg rises (hysteresis ∝ f, eddy ∝ f²), so in practice B_{max} is reduced somewhat at higher f.

- **ii. Doubling N_2.** At the same B_{max}, E_2 doubles. If the window area is unchanged, each turn gets half the copper area ⇒ the thermal current limit halves ⇒ rating **unchanged**. (If the core window could also grow to keep the conductor size, S would double.) The key instead argues that doubling N_2 doubles the MMF for the same current and risks saturation; that applies only if the *primary* voltage is not also changed — the flux in a transformer is set by $V_1/(4.44fN_1)$, not by the load MMF, which is balanced by the primary. We recommend the window-area argument.
- **iii. Doubling A.** For the same voltage, B_{max} halves — no saturation risk but no extra power at the same current. Used properly (keep B_{max} at its limit), the EMF per turn doubles, so the same window can deliver **twice** the power, or the same voltage needs half the turns (thicker wire, more current). Key: rating stays constant at fixed voltage but "it is possible to increase the power rating." Both views are consistent: area raises the *capability*.

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7. MT1 recitation (Dr. Ermiş) and Homework 1

Source. `exams/archive/MT/MT1 Recitation Dr. Ermis.pdf` pp. 1–10 (handwritten, all with solutions). **Keys confirmed**, with two notes below.

7.1 V/f-constant excitation (pp. 1–2)

Given. Coil on a core excited from a variable-voltage, variable-frequency source with V/f constant; r_1, x_1 neglected. Rated 230 V at 50 Hz, minimum 46 V at 10 Hz. Design point $\hat{B}_m$ = 1.6 T. N is blank in the question; the key uses N **= 100**.

Equivalent circuit and meanings. $V = E$ across $G_c \parallel -jB_m$. r = winding (copper) loss, x = leakage flux, G_c = core loss (hysteresis + eddy), B_m = magnetizing (mutual) flux.

ii. $E = V$ = 230 V. (iii) $\hat{\Phi}_m = 230/(4.44 \times 100 \times 50)$ = 10.36 mWb ⇒ $A = \hat{\Phi}_m/1.6$ = 6.475 × 10^{-3} m² ≈ **64.8 cm²** (about 8 cm × 8 cm). (iv) At 10 Hz: $E = 4.44(100)(10)(6.475 \times 10^{-3})(1.6)$ = **46 V** — exactly the minimum supply voltage. Keeping V/f = 230/50 keeps $\hat{B}_m$ at 1.6 T at every frequency; that is the principle of V/f drives and the reason MT 2015 Q3(c) saturates.

7.2 10 kVA, 220/110 V transformer at full load, 0.8 pf lagging (pp. 3–5)

Given. Approximate circuit on the HV side: r_{eq} = 0.25 Ω, x_{eq} = 0.25 Ω, g_c = 0.01 S, b_m = 0.05 S. $I_2 = 10\,000/110$ = 90.9 A ⇒ I_2' = 45.45 A; V_2' = 220 V reference.

a. $\mathbf{V}_1 = 220 + 45.45(0.8 - j0.6)(0.25 + j0.25) = 220 + 45.45(0.35 + j0.05) = 235.9 + j2.27 =$ **$235.9\angle 0.55^\circ$ V**.
b. P_{out} = 8 kW; $P_{cu} = 45.45^2(0.25)$ = 516.5 W; $P_{core} = V_1^2 g_c$ = 556.6 W ⇒ η = 8000/9073 = **88.2 %**.
c. Regulation: the key computes $(236 - 220)/236$ = 6.8 % — it divides by the **no-load** voltage. The course definition (LN10 slide 3, Sen Eq. 2.16, Lesson 20 Eq. 20.1) divides by the **full-load** voltage: $(235.9 - 220)/220$ = **7.24 %**. Use the course definition in exams unless told otherwise (errata E15).

7.3 10 MVA, 13.8/79.7 kV step-up transformer, X = 1.9 Ω (pp. 6–9)

Given. Losses and magnetizing current neglected; rated MVA at rated voltage. $I_1 = I_2' = 10^7/13\,800$ = 724.6 A (HV current 125.5 A).

- 0.8 pf lagging: $\mathbf{V}_1 = 13\,800 + j1.9(724.6)(0.8 - j0.6) = 14\,626 + j1101 = 14\,668\angle 4.31^\circ$ V ⇒ VR = **+6.29 %**. The key rounds $|V_1|$ to 14.7 kV and prints 6.5 % — rounding only.
- 0.8 pf leading: $\mathbf{V}_1 = 12\,974 + j1101 = 13\,021\angle 4.85^\circ$ V ⇒ VR = **−5.65 %** (key −5.7 %). With a capacitive load the no-load voltage is *lower* than the full-load voltage: negative regulation, as in Lesson 20 §2.2.

7.4 Homework 1 (due 4 Nov 2014), three-leg core (p. 10)

Given. Coil on leg 1; legs 2 (centre, ℓ_2 = 0.1 m) and 3 (ℓ_3 = 0.3 m) in parallel; ℓ_1 = 0.3 m; all areas A = 9 × 10⁻⁴ m². B–H table: (0.5 T, 50 A/m), (0.9, 150), (1.0, 200), (1.4, 750). Find I for B_3 = 0.5 T and the leg fluxes. N **is not given** (gap) — we give the MMF NI.

- Leg 3: B_3 = 0.5 T ⇒ H_3 = 50 A/m ⇒ $H_3\ell_3$ = 15 A·t. Legs 2 and 3 share the same MMF: $H_2 = 15/0.1$ = 150 A/m ⇒ B_2 = 0.9 T.
- Fluxes: Φ_3 = 0.45 mWb, Φ_2 = 0.81 mWb, $\Phi_1 = \Phi_2 + \Phi_3$ = **1.26 mWb** ⇒ B_1 = 1.4 T ⇒ H_1 = 750 A/m ⇒ $H_1\ell_1$ = 225 A·t.
- $NI = 225 + 15$ = **240 A·t** (e.g. 2.4 A for 100 turns). Same method as Lesson 12, Examples 12.4 and 12.6 (flux given, nonlinear iron, parallel paths). Leg 1 works near saturation and takes 94 % of the MMF.
